

ANIKET

NETWORK FORENSICS

MODULE – 8

Aniket Sunil Pagare

✦ 1. What is Network Forensics?

Definition:

Network Forensics is the **process of capturing, recording, and analyzing network events** to investigate **security incidents**, detect **malicious activity**, and gather **digital evidence** for legal or investigative purposes.

It focuses on **live network traffic** (packets, flows, logs) rather than stored data on a device.

✦ 2. Objectives of Network Forensics

- **Incident Response** → Identify & respond to security breaches in real time.
 - **Evidence Collection** → Collect admissible digital evidence for legal cases.
 - **Intrusion Detection** → Identify malware, unauthorized access, and suspicious behavior.
 - **Traffic Analysis** → Understand communication patterns and anomalies.
 - **Data Recovery** → Recover stolen or lost information from intercepted packets.
-

✦ 3. Key Principles in Network Forensics

1. **Chain of Custody** – Document every step of evidence handling.
2. **Data Integrity** – Ensure captured packets are unmodified (e.g., using hashes like MD5/SHA256).
3. **Minimal Interference** – Capture traffic passively without affecting operations.
4. **Time Synchronization** – Ensure all logs and captures use synchronized timestamps (NTP).
5. **Legal Compliance** – Follow local laws on monitoring and privacy.

📌 4. Network Forensics Process

Step 1: Preparation

- Deploy monitoring points (SPAN ports, TAP devices).
- Install forensic tools (Wireshark, tcpdump, Zeek, etc.).
- Configure NTP for accurate timestamps.

Step 2: Data Acquisition

- Capture **full packet data** or **NetFlow records**.
- Save in standard formats (PCAP, PCAPNG).
- Example:

```
sudo tcpdump -i eth0 -w capture.pcap
```

Step 3: Data Preservation

- Store packet captures securely.
- Generate a hash for verification:

```
sha256sum capture.pcap > capture_hash.txt
```

Step 4: Data Analysis

- Filter suspicious IPs/ports in Wireshark.
- Identify protocols and abnormal behavior.
- Correlate with firewall, IDS/IPS logs.

Step 5: Reporting

- Create a detailed incident report.
- Include timeline, attacker IPs, methods used, and recommendations.

📌 5. Data Sources in Network Forensics

- **Packet captures (PCAP)**
- **Firewall logs**

- **IDS/IPS alerts (Snort, Suricata)**
 - **DNS logs**
 - **VPN logs**
 - **Syslogs from network devices**
-

✦ **6. Types of Network Forensics**

1. **Catch-it-as-you-can** – Capture all packets for later analysis (high storage cost).
 2. **Stop-look-and-listen** – Analyze packets in real time, save only important ones.
-

✦ **7. Common Protocols Investigated**

- HTTP / HTTPS
 - FTP / SFTP
 - SMTP / IMAP / POP3 (emails)
 - DNS queries
 - SMB (file sharing)
 - SSH / Telnet
 - ICMP (pings)
-

✦ **8. Common Attacks Detected via Network Forensics**

- **DDoS Attacks**
- **ARP Spoofing / MITM**
- **Port Scanning**
- **Malware C2 Communications**
- **Data Exfiltration**
- **SQL Injection via HTTP**

- **Phishing Emails**

🔥 9. Popular Network Forensic Tools

Tool	Purpose
Wireshark	Packet capture & protocol analysis
tcpdump	CLI packet capture
Zeek (Bro)	Network traffic analysis & scripting
NetworkMiner	Extract files, images, credentials from PCAP
Xplico	Reconstruct web/email/VoIP sessions
Snort / Suricata	IDS/IPS detection
Moloch (Arkime)	Large-scale packet capture indexing
PcapXray	Visualize network traffic flows
NetFlow Analyzer	Flow-based traffic analysis

🔥 10. Legal Considerations

- Monitoring without consent may be **illegal** in many countries.
- Always get **proper authorization** before capturing traffic.
- Keep **evidence handling logs** for court admissibility.

🔥 11. Challenges in Network Forensics

- Encrypted traffic (HTTPS, VPN) makes deep inspection harder.
 - High bandwidth → storage overload.
 - Evasion techniques like tunneling or steganography.
 - Distributed attacks from multiple sources.
-

Lab-1 – Identify And Investigate an Logs Attacks Using Splunk

1. What is Splunk?

Definition:

Splunk is a **log and data analysis platform** that collects, indexes, and analyzes **machine-generated data** from:

- Servers
- Applications
- Network devices
- Security tools
- IoT devices

Key Strength:

It turns **raw log data** into **searchable, visualized, and actionable intelligence** in real time.

2. Why Use Splunk?

- Centralized log management
 - Real-time monitoring
 - Security Information and Event Management (SIEM)
 - Incident detection and response
 - Compliance and audit trails
 - Forensics investigation
-

3. Splunk Components

Splunk has three main components:

Component	Role
Forwarder	Collects data from source systems and sends it to the indexer.

Component	Role
Indexer	Stores and indexes incoming data, making it searchable.
Search Head	User interface for searching, analyzing, and visualizing data.

5. Splunk Editions

- **Splunk Enterprise** – Full version for large-scale environments.
- **Splunk Cloud** – Cloud-based version.
- **Splunk Free** – Limited to 500MB/day indexing.
- **Splunk Light** – Lightweight for small businesses.

6. Splunk Data Sources

Splunk can ingest:

- Syslog data (Linux, network devices)
- Windows Event Logs
- Application logs (Apache, Nginx)
- Security logs (Firewall, IDS/IPS)
- Cloud logs (AWS CloudTrail, Azure)
- Database logs
- IoT telemetry

8. Splunk User Interface

- **Search & Reporting App** – Core search and analysis.
- **Data Summary** – Overview of indexed sources.
- **Dashboards** – Visual representation of search results.
- **Alerts** – Automated notifications based on search conditions.

14. Splunk for Incident Response

Process:

1. Ingest logs from all sources.
 2. Search for anomalies.
 3. Create alerts and dashboards.
 4. Correlate events to find root cause.
 5. Export evidence with timestamps.
-

16. Splunk Limitations

- Expensive for large-scale environments.
 - Needs proper configuration to avoid performance issues.
 - Learning SPL can take time.
-

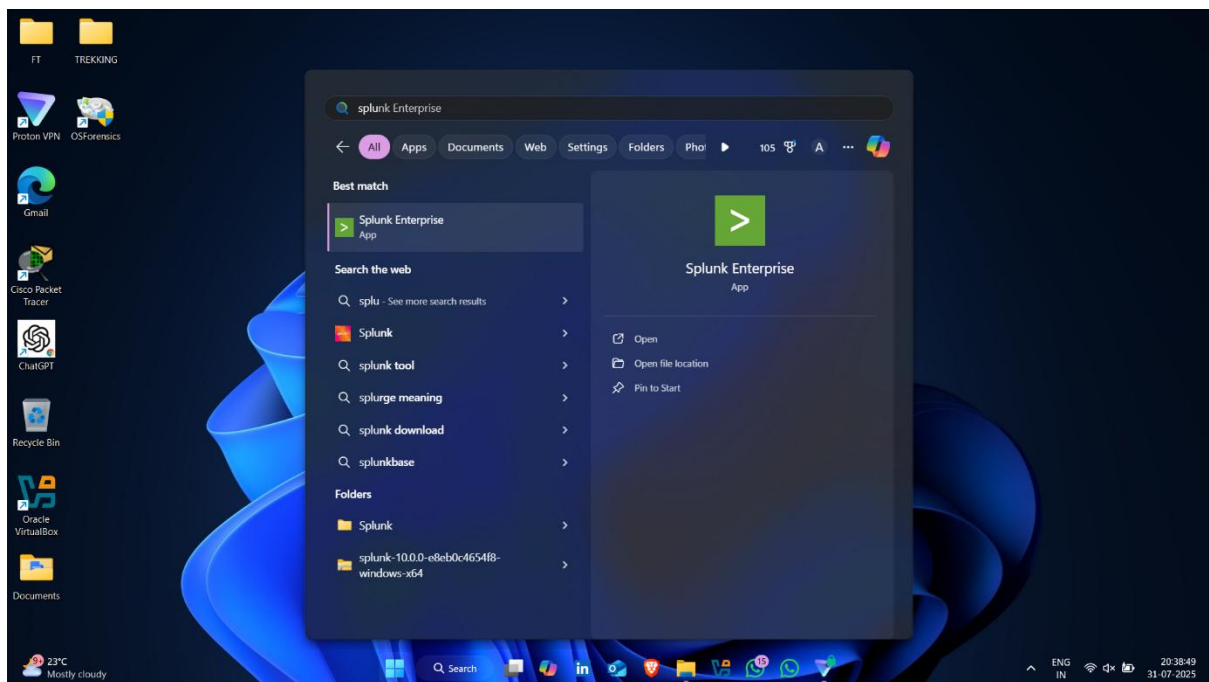
17. Splunk Alternatives

- **ELK Stack** (Elasticsearch, Logstash, Kibana)
 - **Graylog**
 - **SIEMonster**
 - **Wazuh**
-

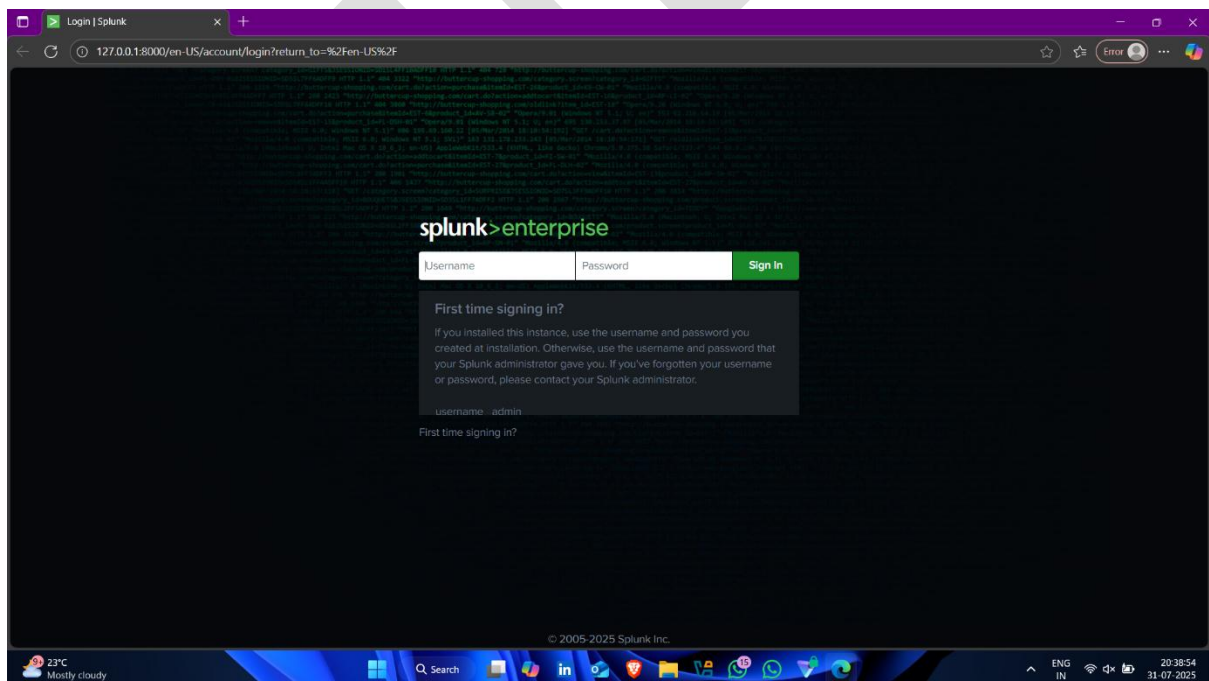
Download Link -:

How to use it :-

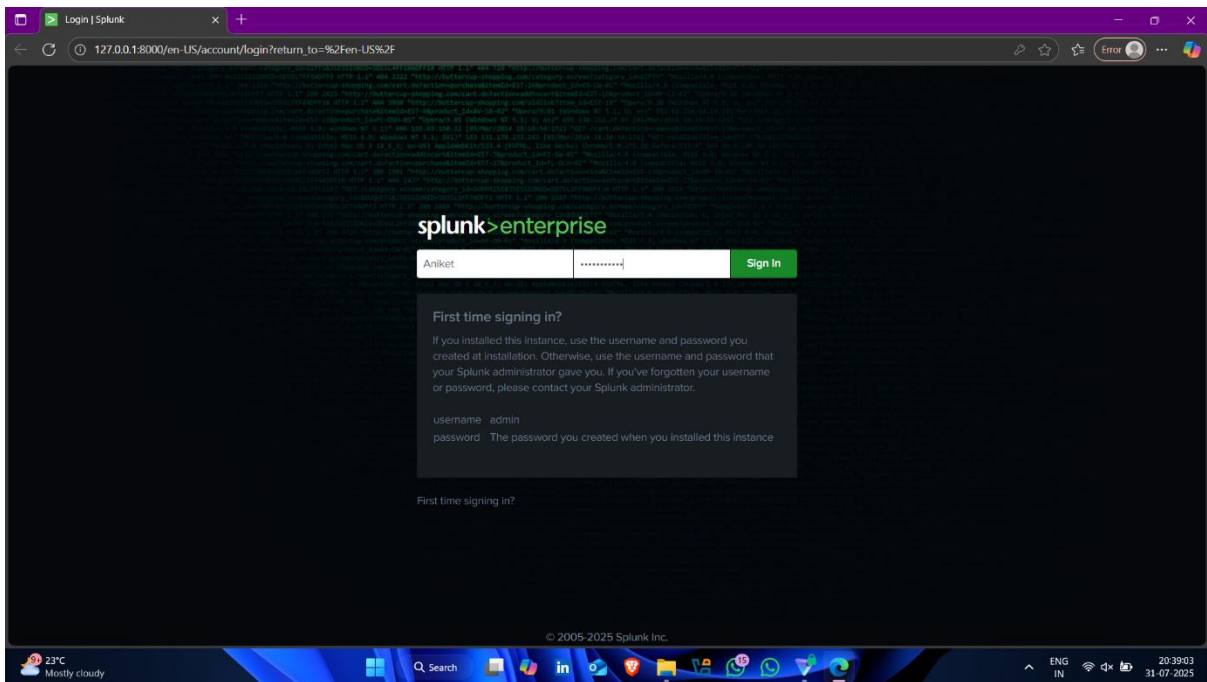
- Open Splunk Enterprise



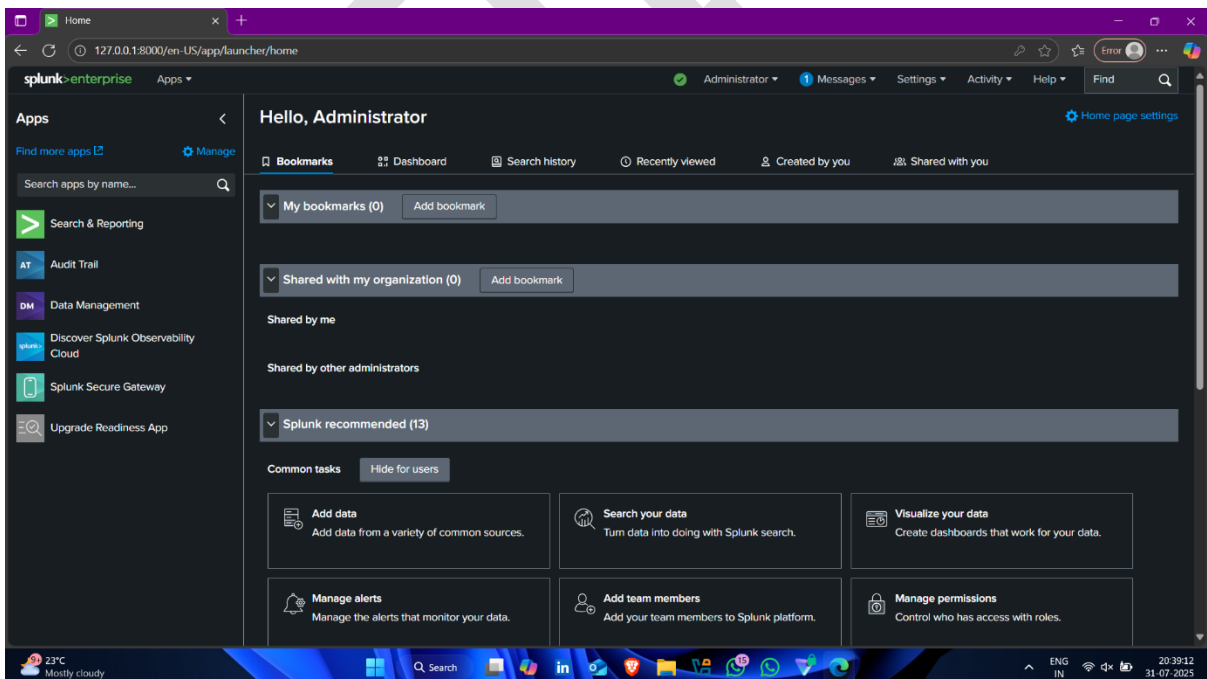
- Enter Username and Password



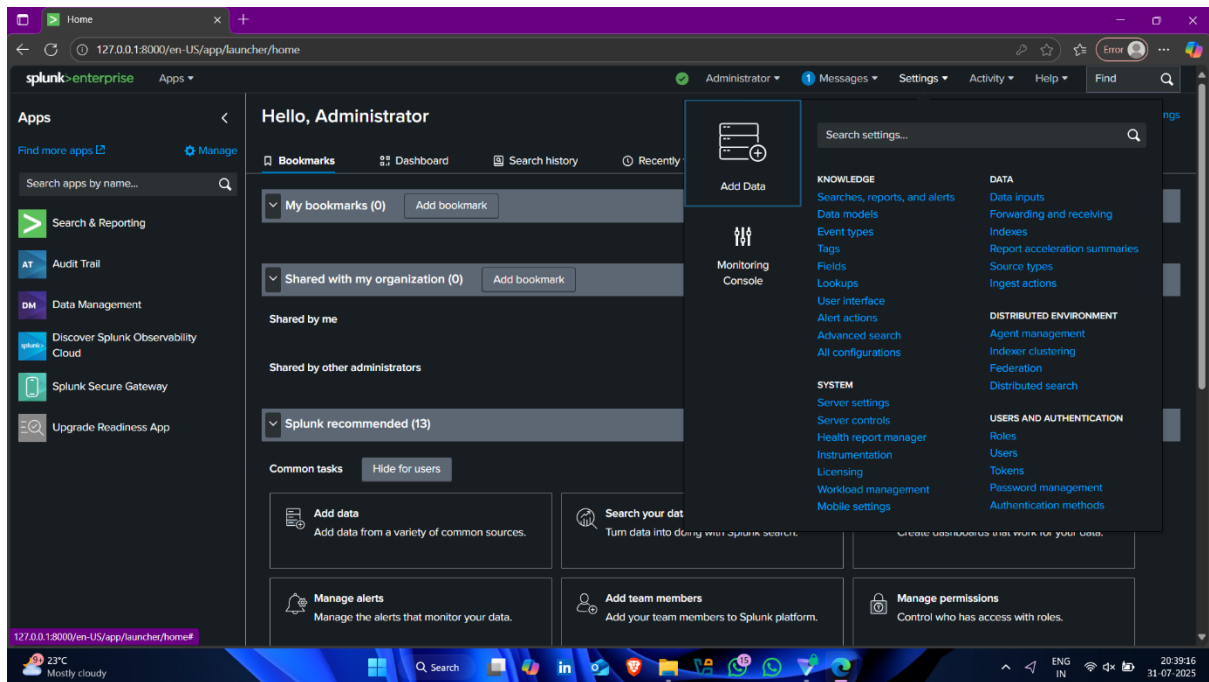
- Click on **Sign in**



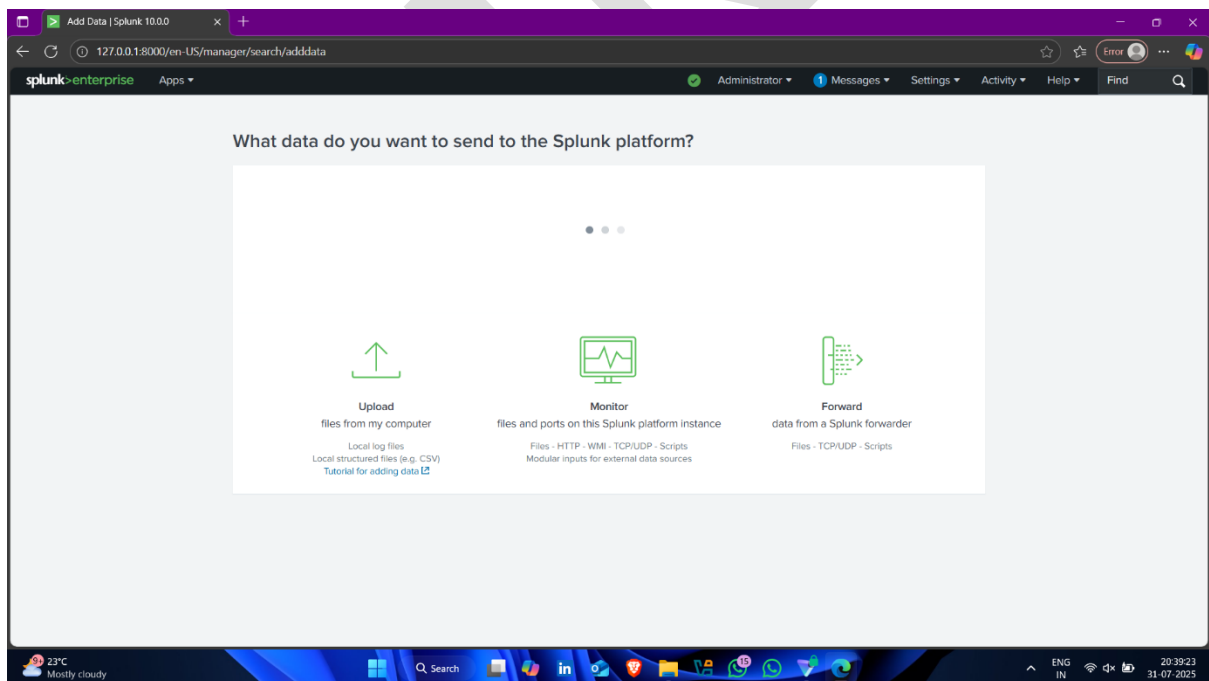
- Splunk Interface ✓👉
- Click on **Settings**



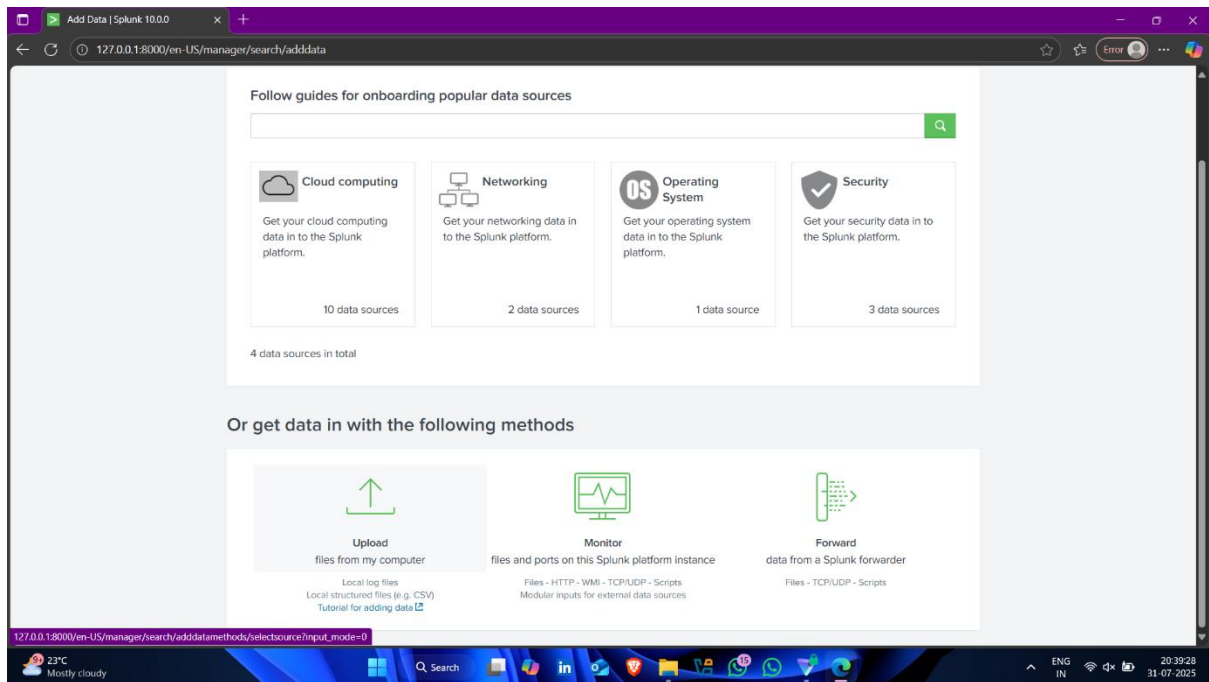
- Click on **Add Data**



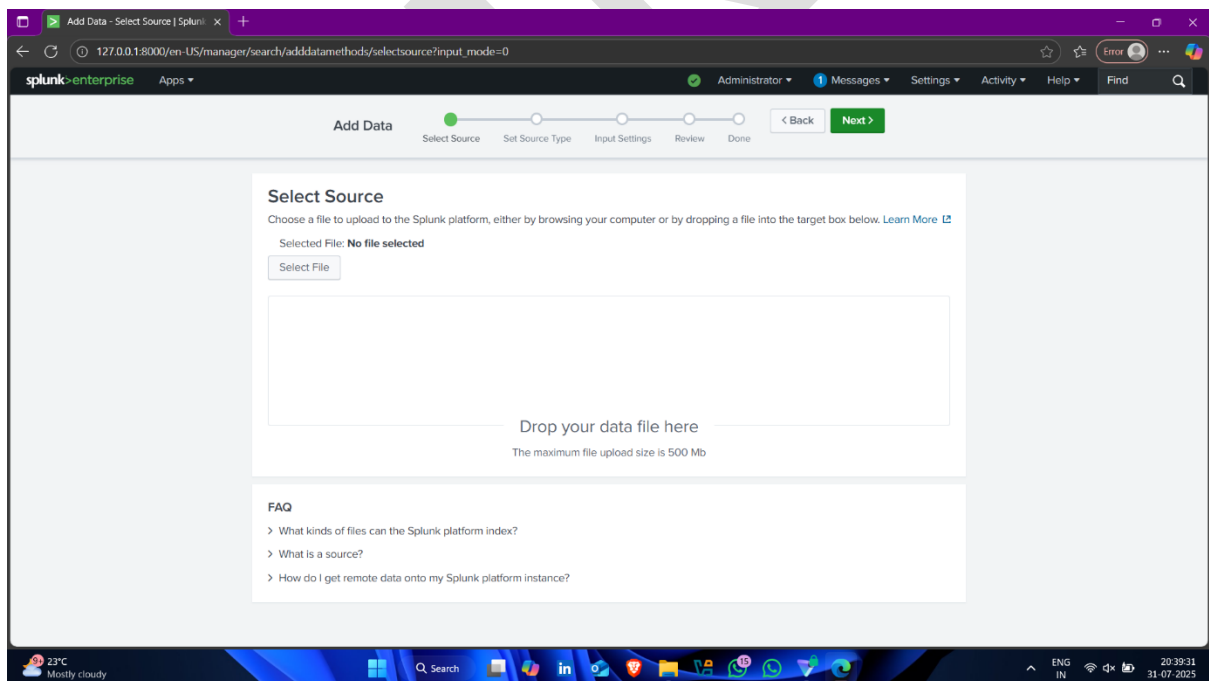
- Here , many Options are available

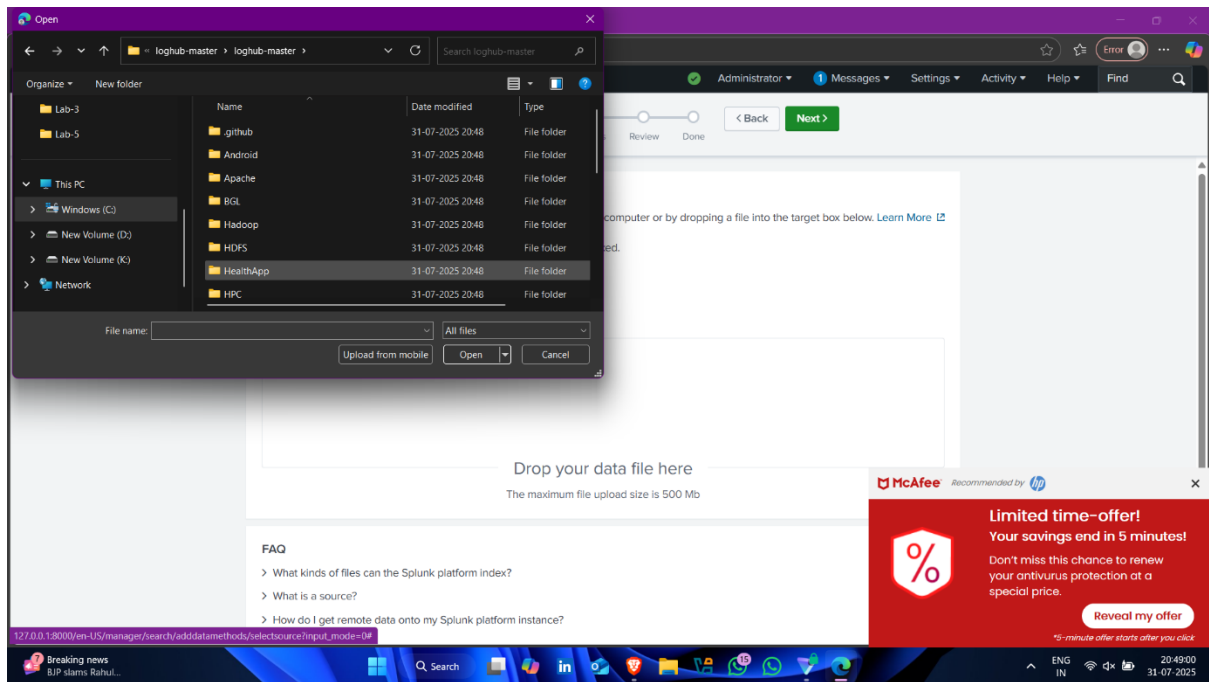


- Click on **Upload** option

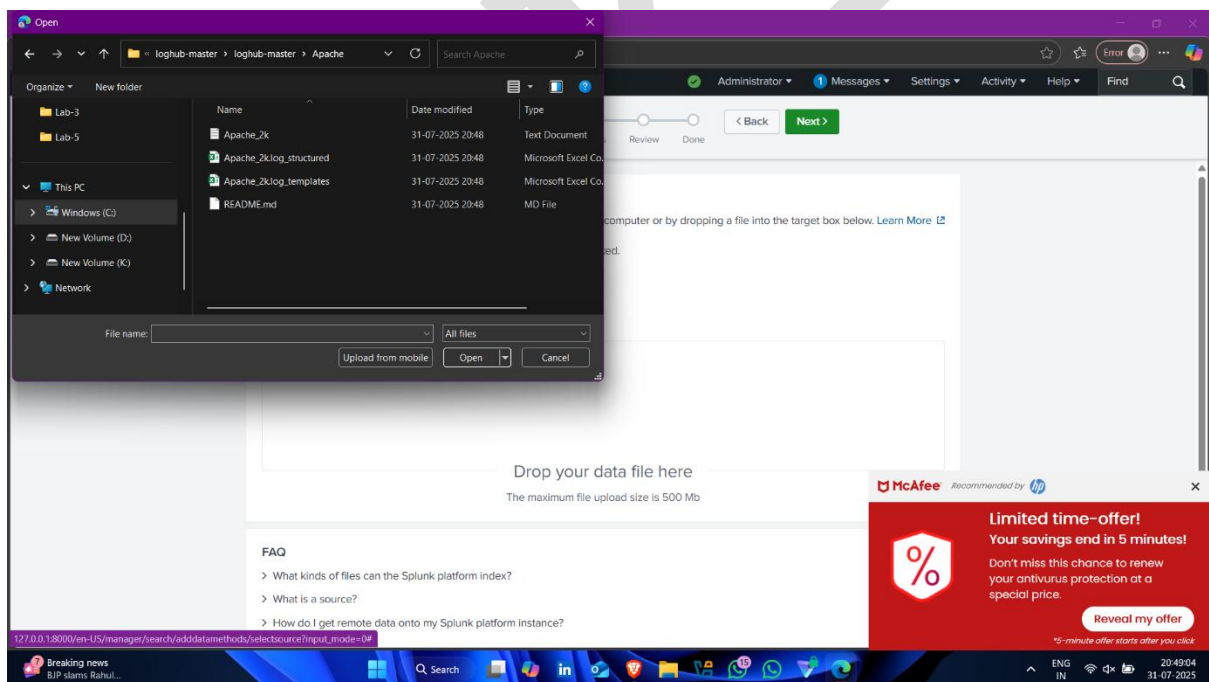


- Now Click on **Select File** Option

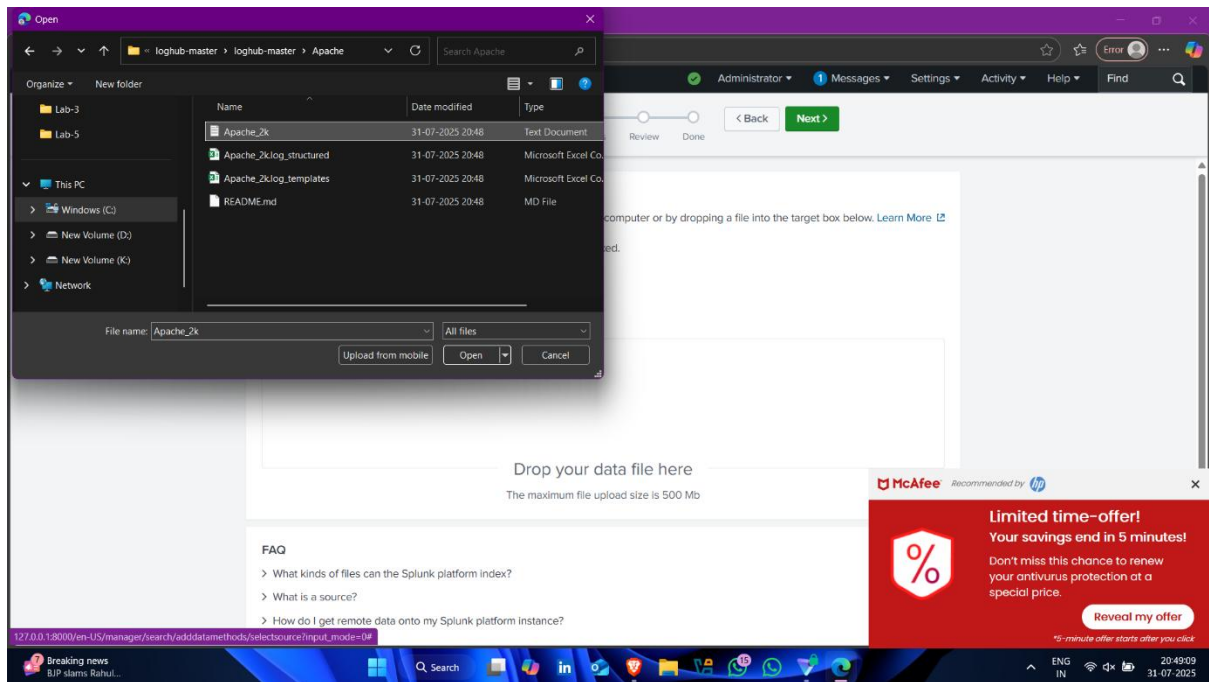




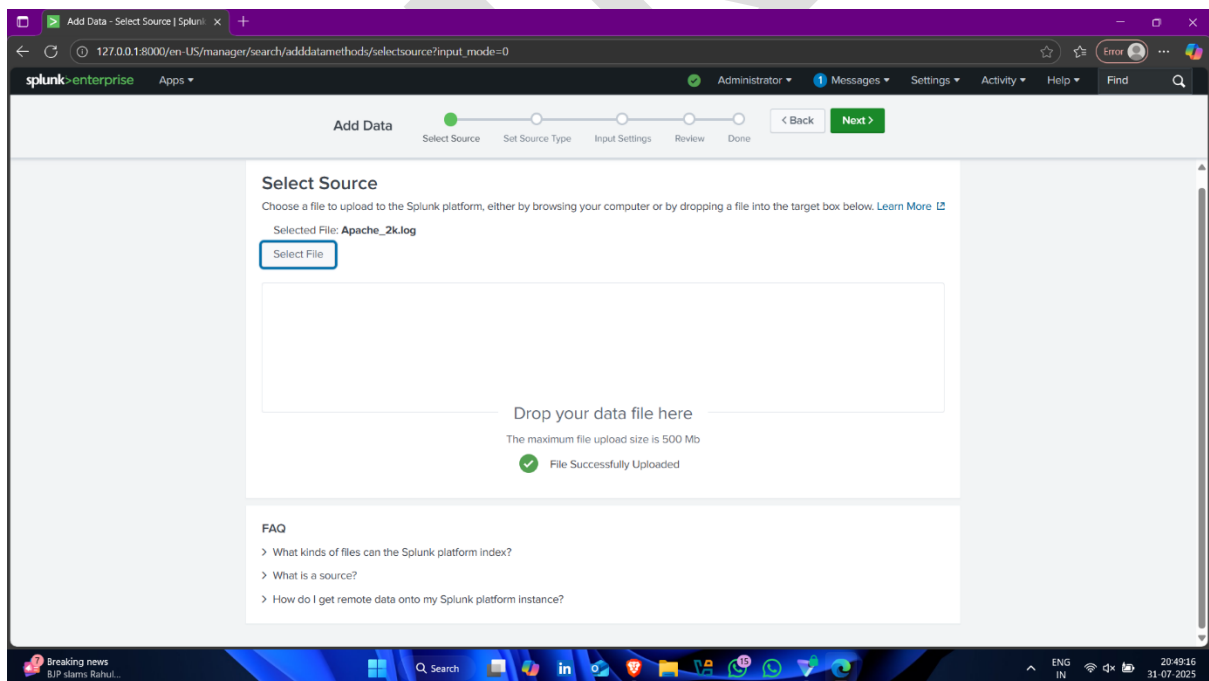
- Select log file



- Click on **Open**



- Log file Uploaded , click on **next**



- Click on Next

Add Data Progress: Select Source, Set Source Type, Input Settings, Review, Done < Back Next >

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **Apache_2k.log** View Event Summary

Source type: default Save As

	Time	Event
1	7/31/25 4:47:44.000 AM	[Sun Dec 04 04:47:44 2005] [notice] workerEnv.init() ok /etc/httpd/conf/workers2.properties [Sun Dec 04 04:47:44 2005] [error] mod_jk child workerEnv in error state 6 [Sun Dec 04 04:51:08 2005] [notice] jk2_init() Found child 6725 in scoreboard slot 10 [Sun Dec 04 04:51:09 2005] [notice] jk2_init() Found child 6726 in scoreboard slot 8 [Sun Dec 04 04:51:09 2005] [notice] jk2_init() Found child 6728 in scoreboard slot 6 Show all 257 lines
2	7/31/25 6:20:30.000 AM	[Sun Dec 04 06:20:30 2005] [notice] jk2_init() Found child 32466 in scoreboard slot 11
3	7/31/25 6:20:30.000 AM	[Sun Dec 04 06:20:30 2005] [notice] jk2_init() Found child 32457 in scoreboard slot 6
4	7/31/25 6:20:44.000 AM	[Sun Dec 04 06:20:44 2005] [notice] workerEnv.init() ok /etc/httpd/conf/workers2.properties
5	7/31/25 6:20:44.000 AM	[Sun Dec 04 06:20:44 2005] [notice] workerEnv.init() ok /etc/httpd/conf/workers2.properties
6	7/31/25 6:20:44.000 AM	[Sun Dec 04 06:20:44 2005] [notice] workerEnv.init() ok /etc/httpd/conf/workers2.properties

Add Data Progress: Select Source, Set Source Type, Input Settings, Review, Done < Back Next >

Set Source Type

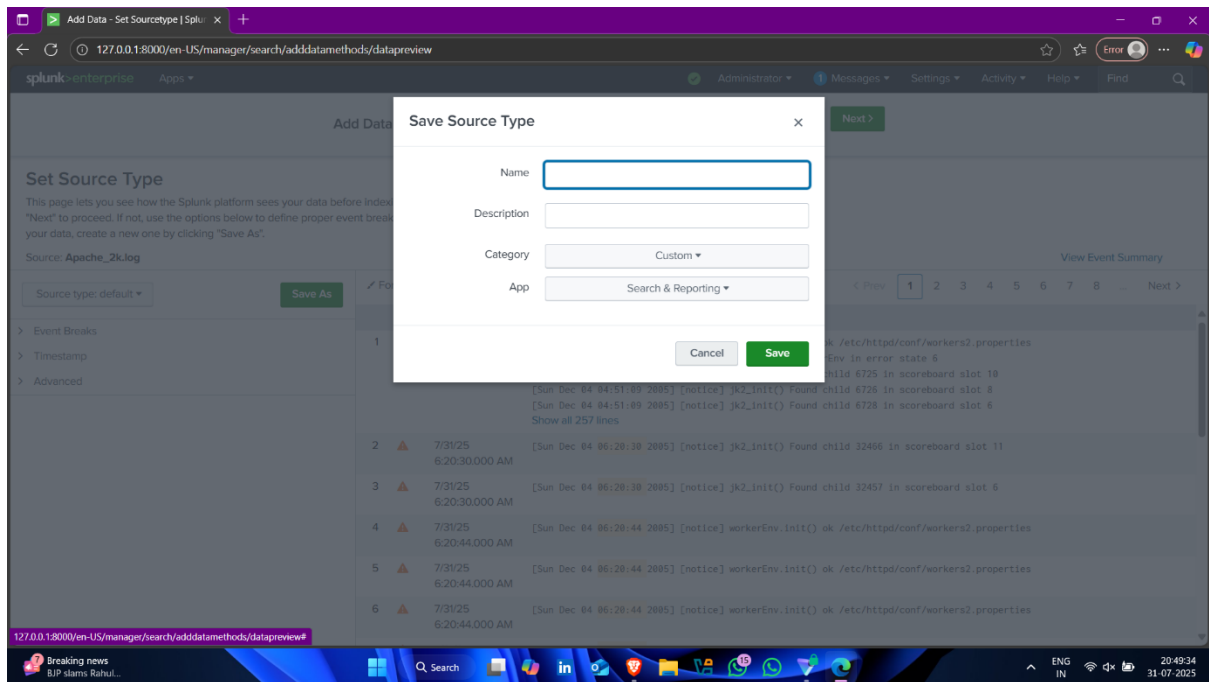
This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **Apache_2k.log** View Event Summary

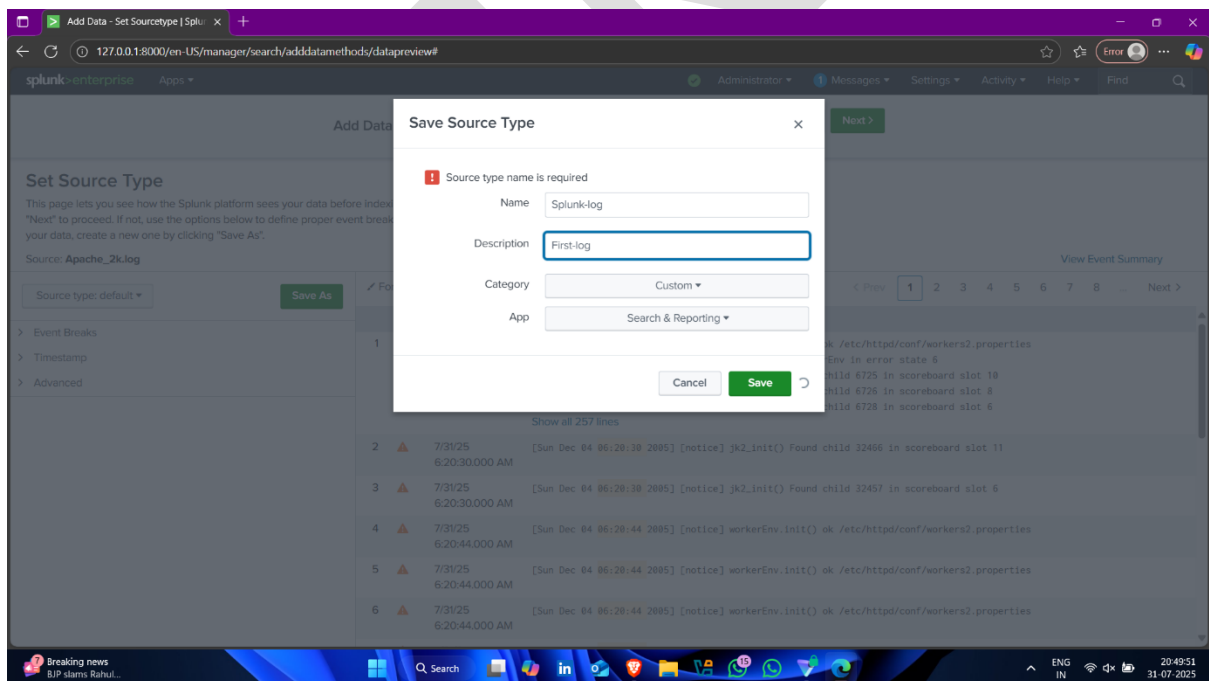
Source type: default Save As

	Time	Event
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2	7/31/25 6:20:30.000 AM	[Sun Dec 04 06:20:30 2005] [notice] jk2_init() Found child 32466 in scoreboard slot 11
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6	7/31/25 6:20:44.000 AM	[Sun Dec 04 06:20:44 2005] [notice] workerEnv.init() ok /etc/httpd/conf/workers2.properties

- Enter Log File name



- Click on **save** option



- Click on **review**

Add Data | Select Source | Set Source Type | **Input Settings** | Review | Done | < Back | Review >

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Constant value ☒ Regular expression on path ☐ Segment in path ☐

Host field value:

Index

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Index: [Create a new index](#)

FAQ

- > How do indexes work?
- > How do I know when to create or use multiple indexes?

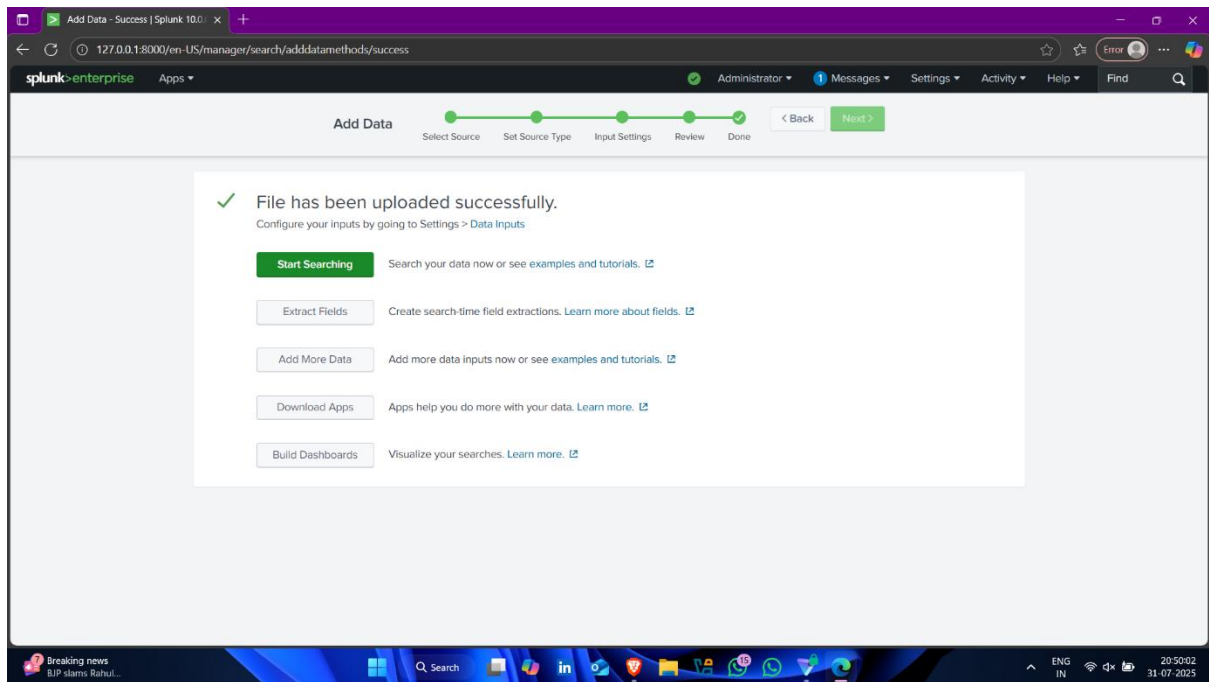
- Click on **submit**

Add Data | Select Source | Set Source Type | Input Settings | **Review** | Done | < Back | Submit >

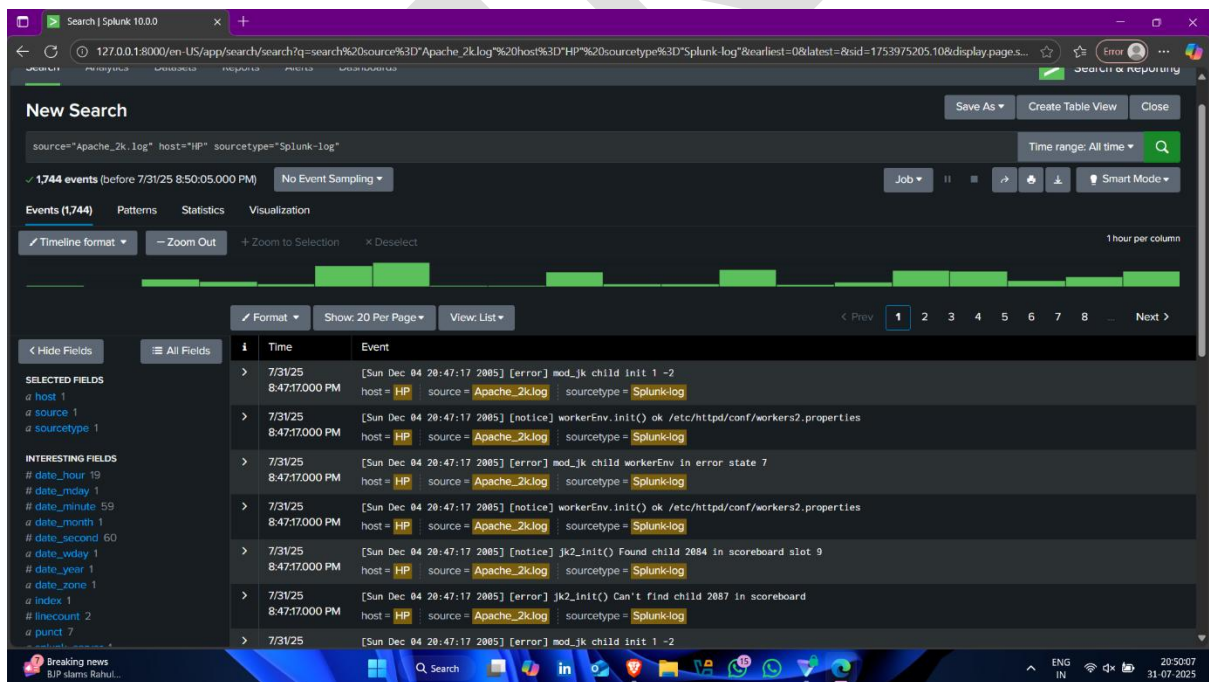
Review

Input Type Uploaded File
File Name Apache_2k.log
Source Type Splunk-log
Host HP
Index Default

- Now click on **Start Searching**



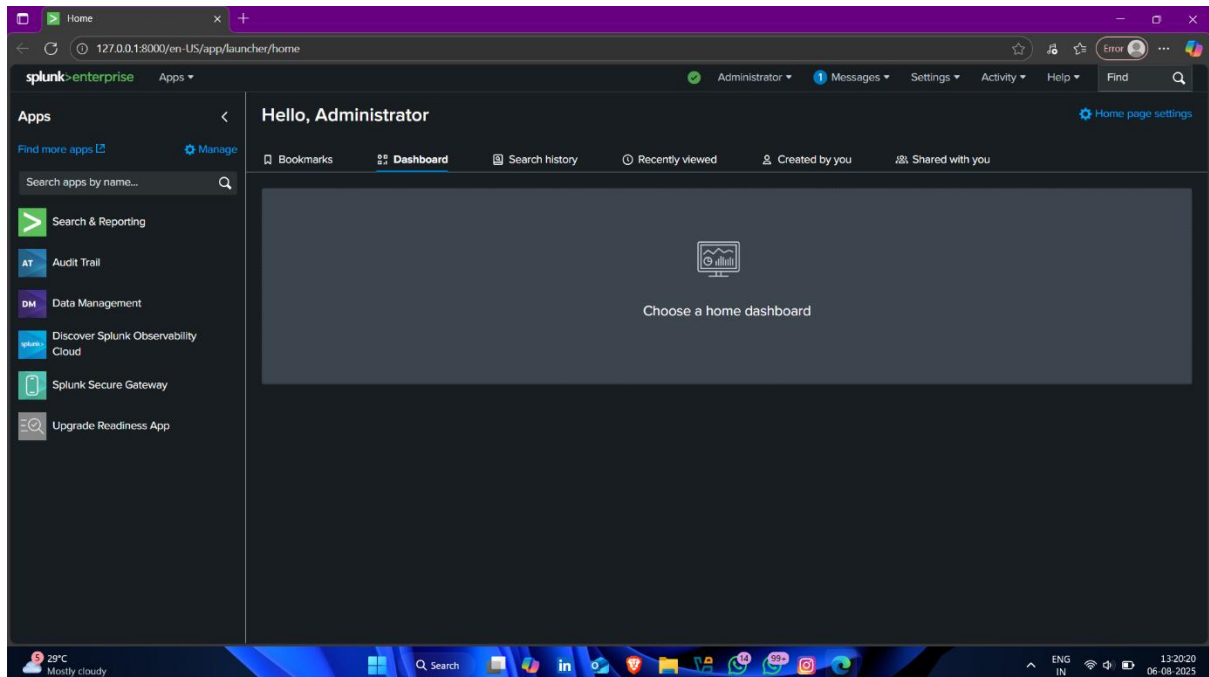
- Logs 🖱️✅



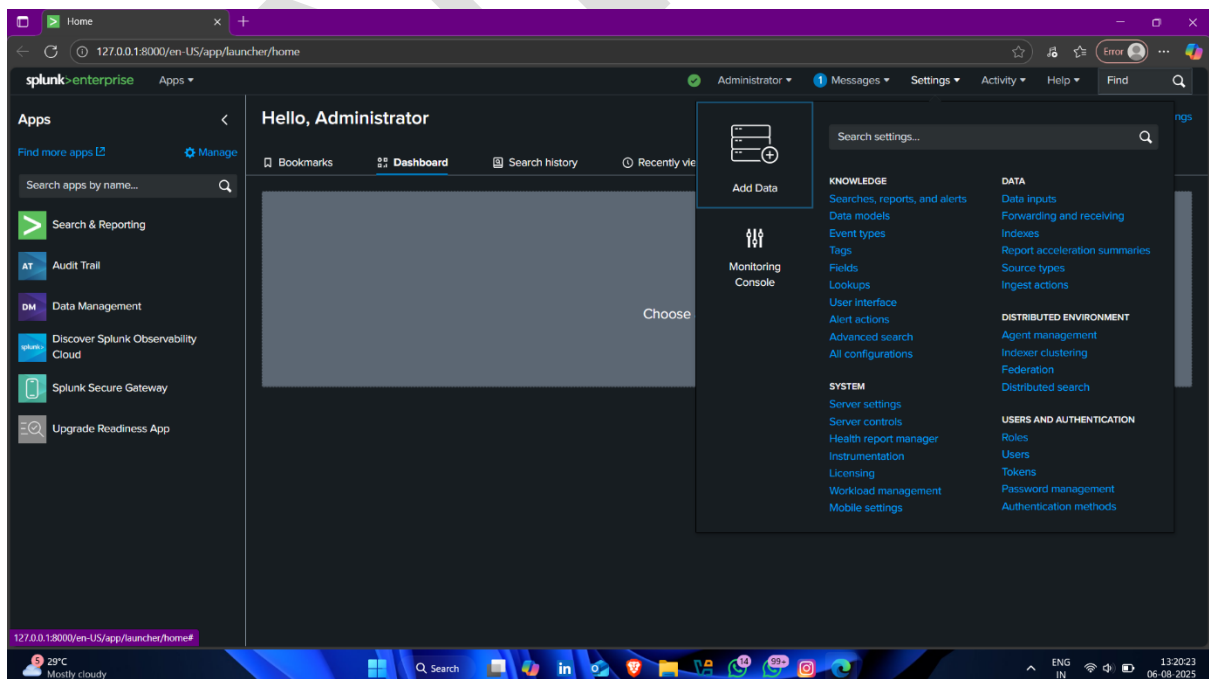
FTP Brute Force Attack Log Analysis Using Splunk

How to do it -:

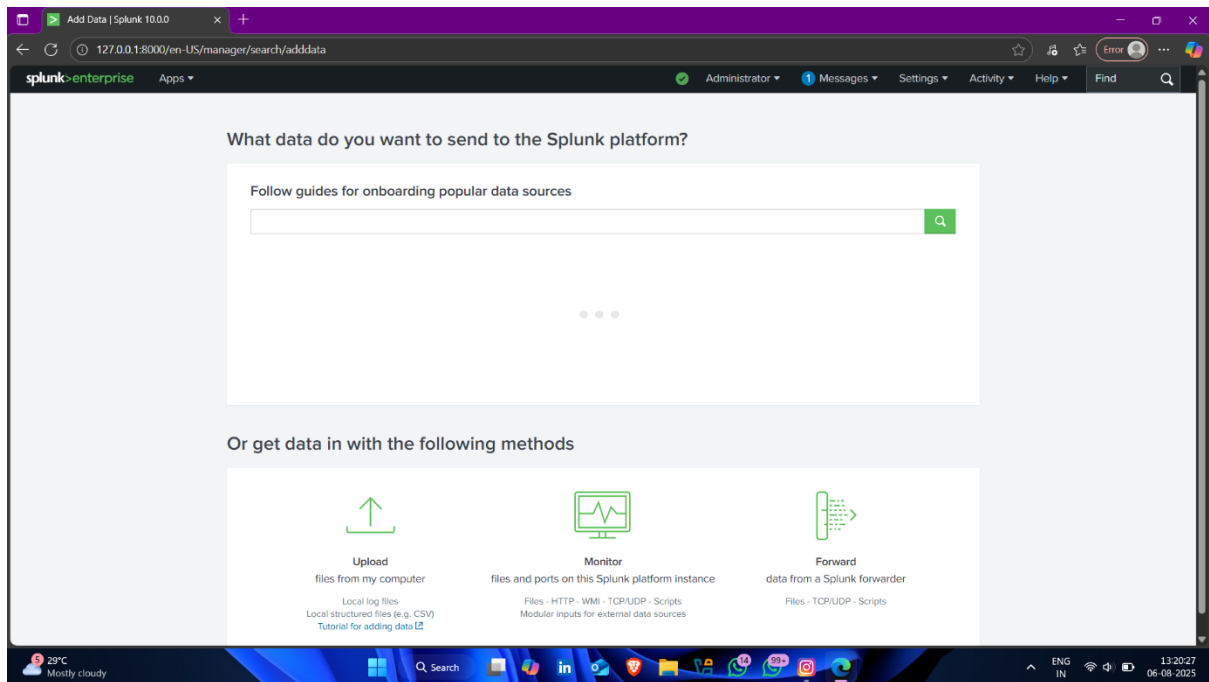
- Open Splunk Enterprise



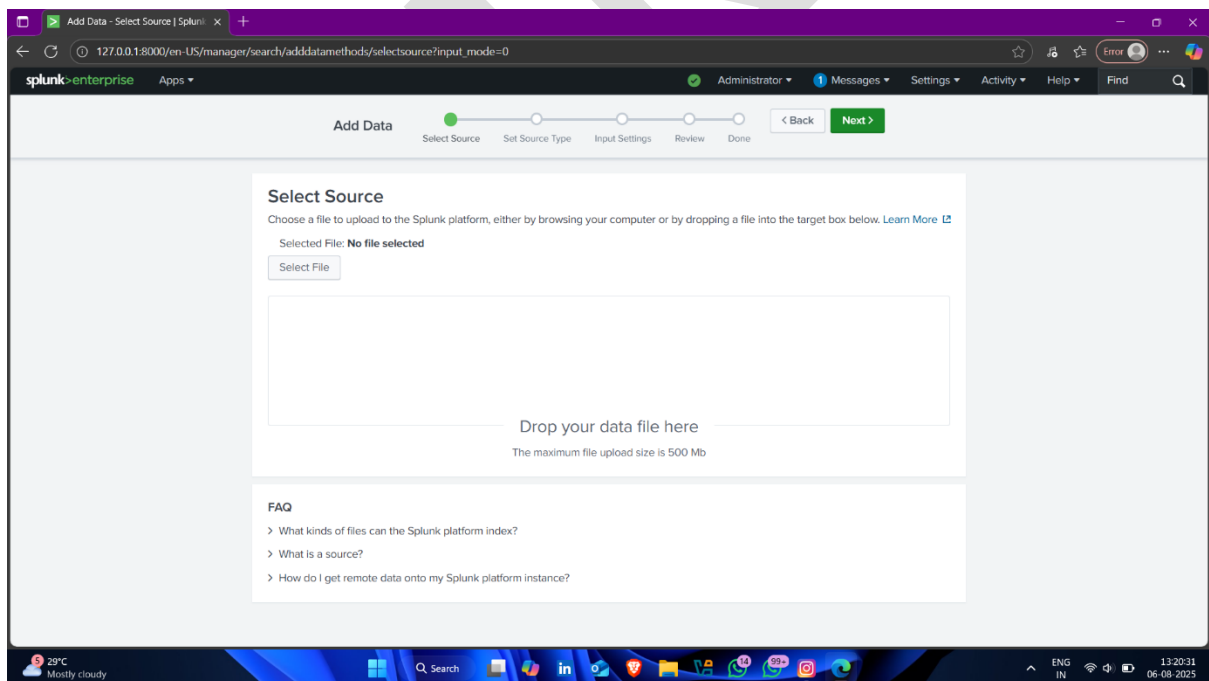
- Click on **Settings** and then click on **Add Data**



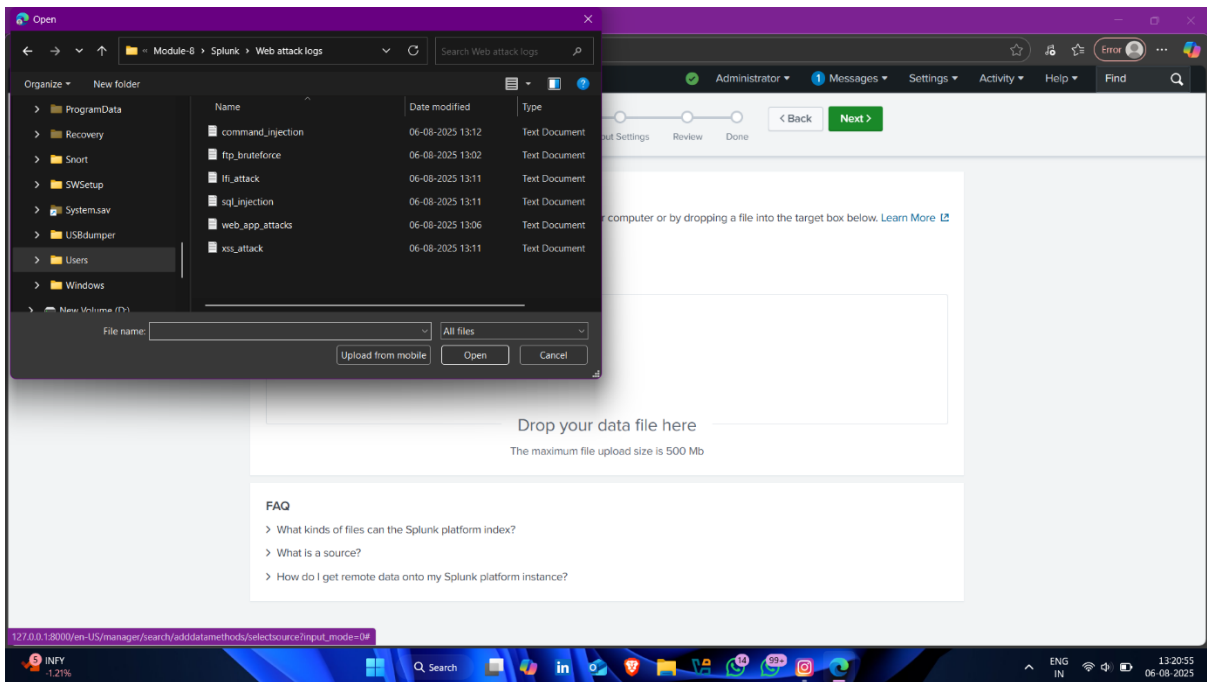
- Click on **Upload**



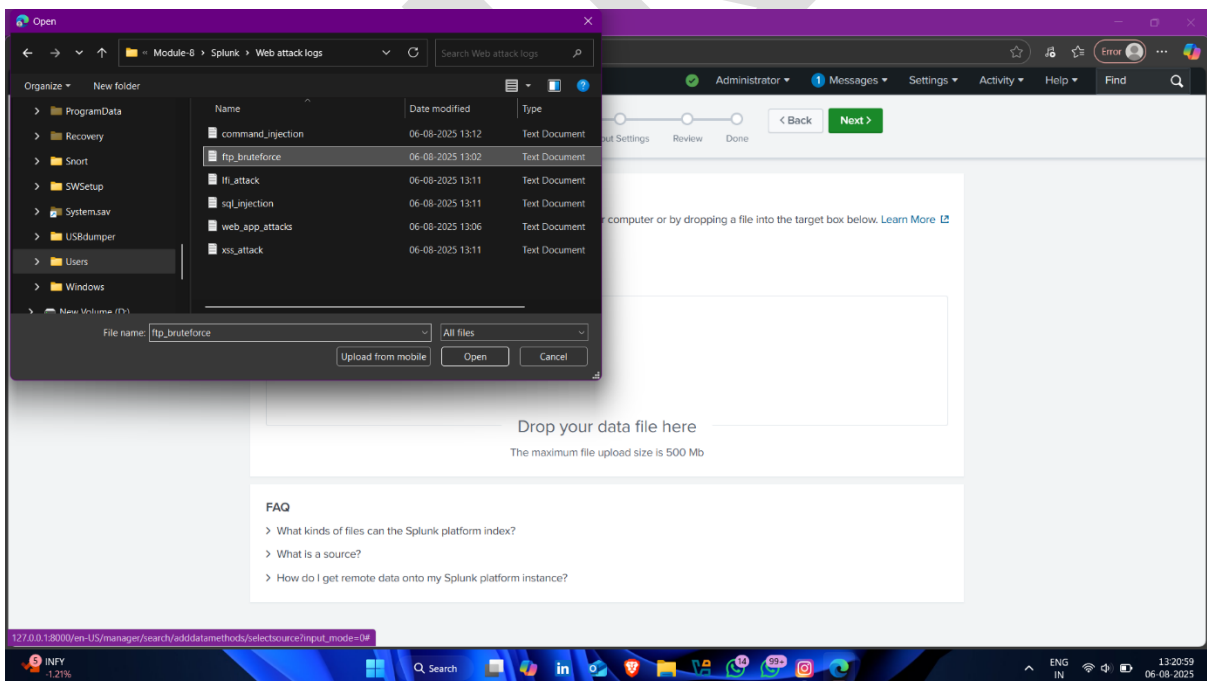
- click on **Select file**



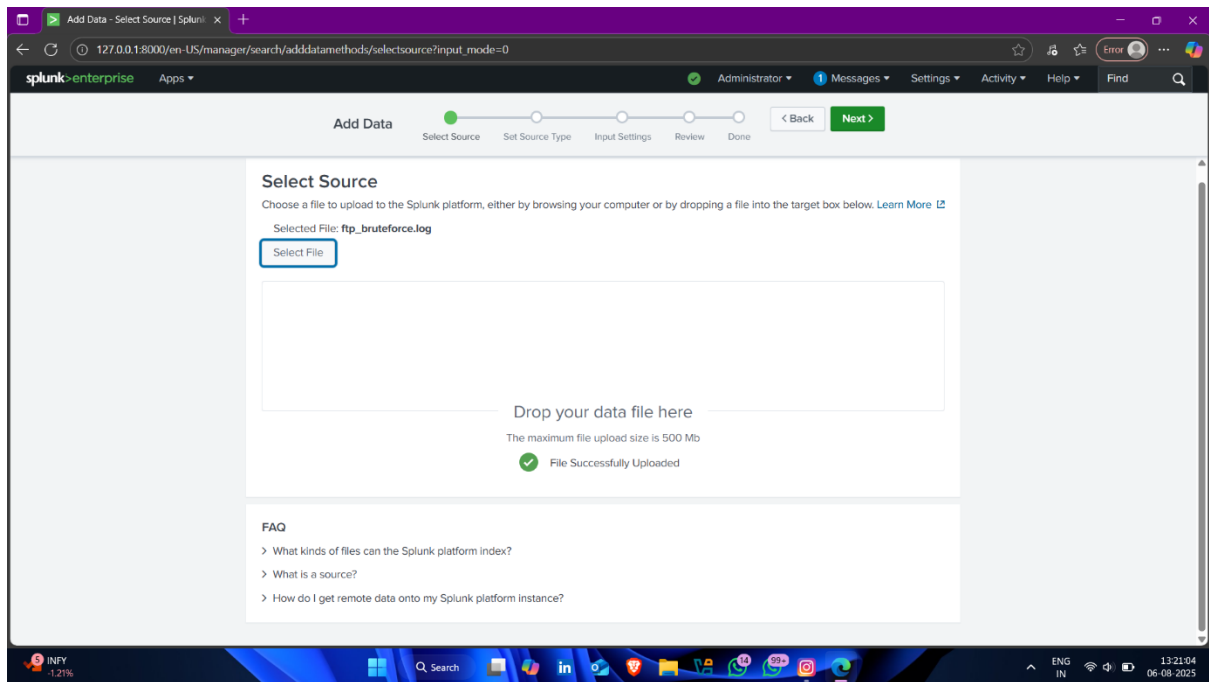
- log files 🖱️



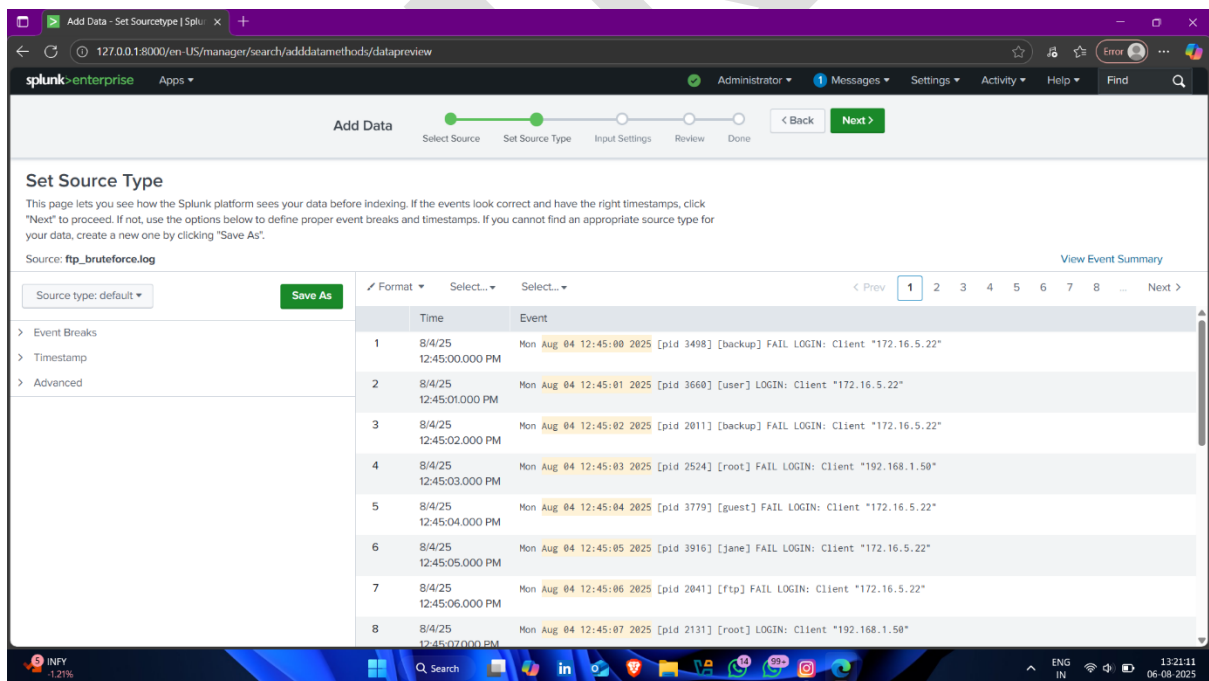
- select Ftp-brute-force-log file and click on open



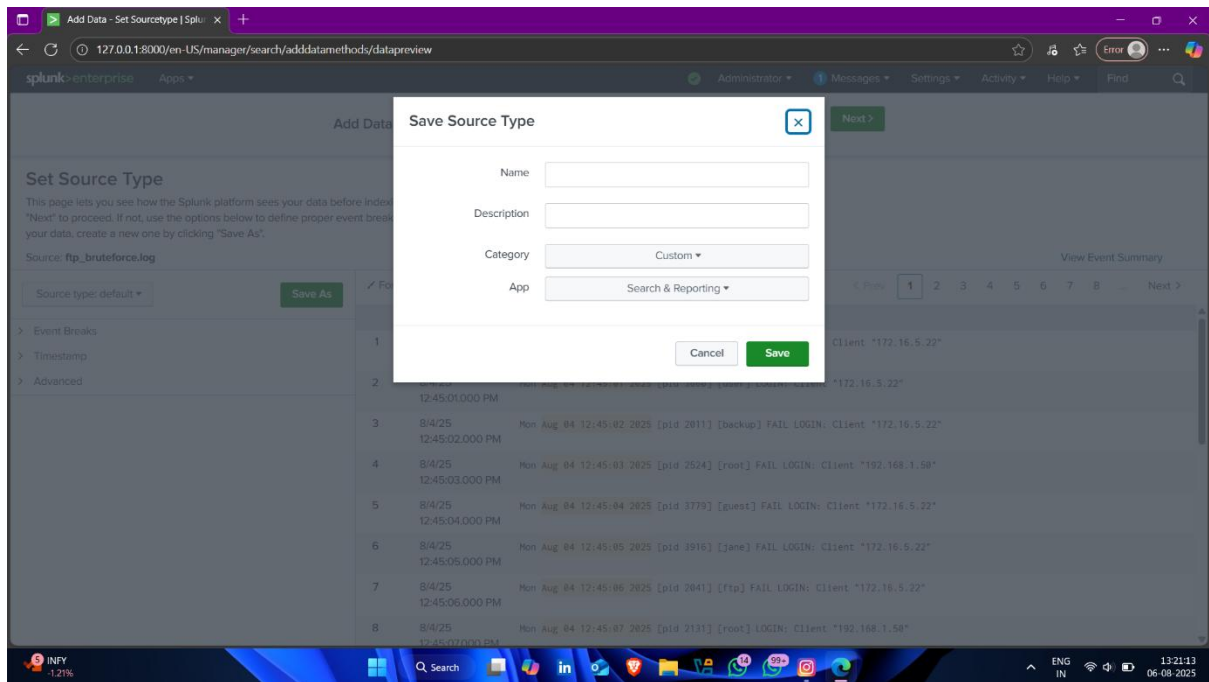
- Log File Uploaded , click on **next**



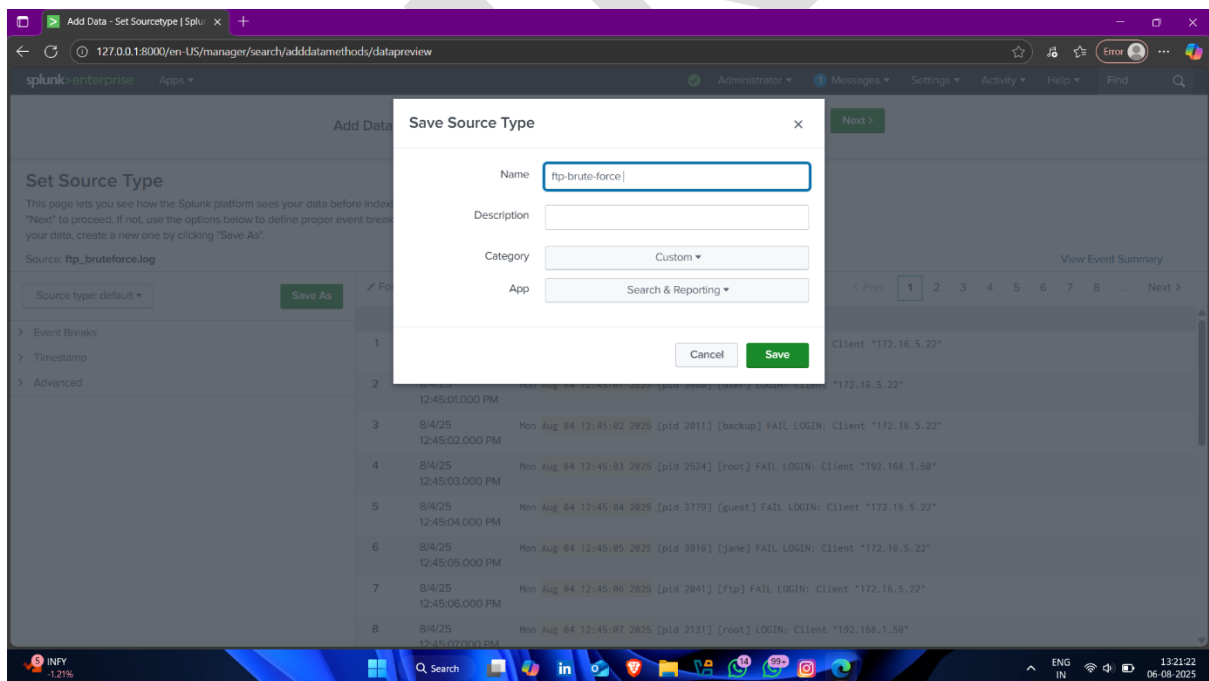
- Click on **next**



- Enter name for source file



- After giving the name , Click on **Save**



- Click on **Review**

Add Data | Select Source | Set Source Type | **Input Settings** | Review | Done

Input Settings
Optionally set additional input parameters for this data input as follows:

Host
When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Host field value:

Index
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Index: [Create a new index](#)

FAQ
> How do indexes work?
> How do I know when to create or use multiple indexes?

[Back](#) [Review](#)

- Click on **Submit**

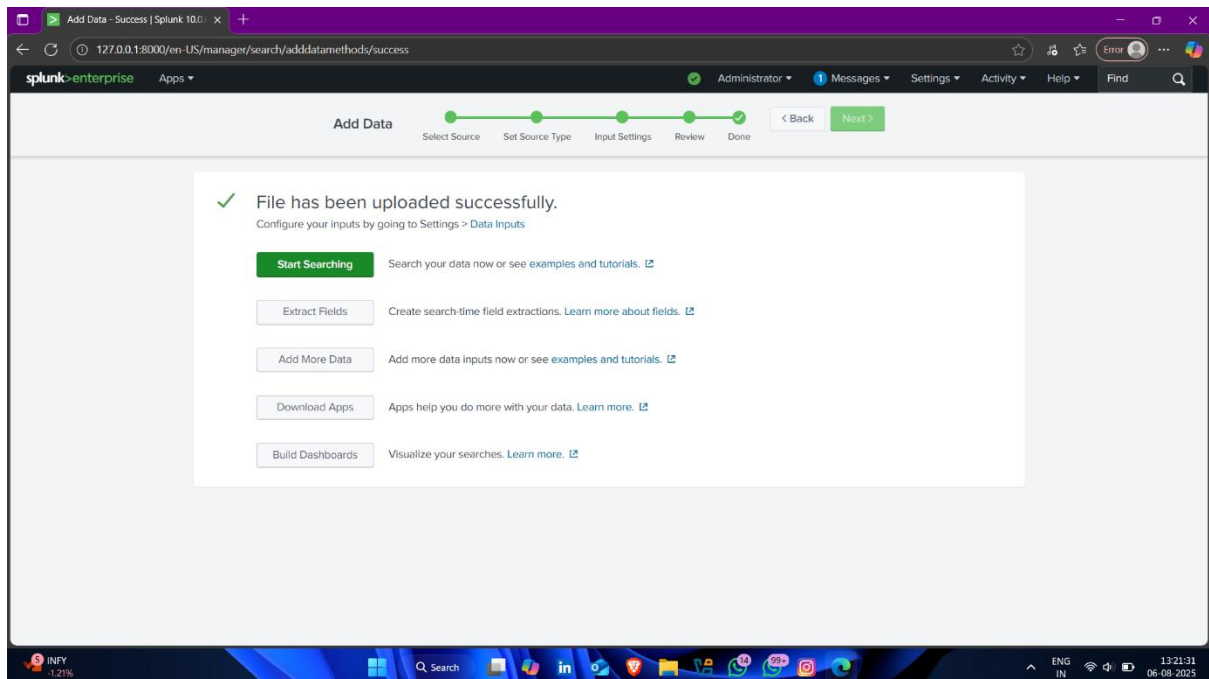
Add Data | Select Source | Set Source Type | Input Settings | **Review** | Done

Review

Input Type Uploaded File
File Name ftp_bruteforce.log
Source Type ftp_brute-force
Host HP
Index Default

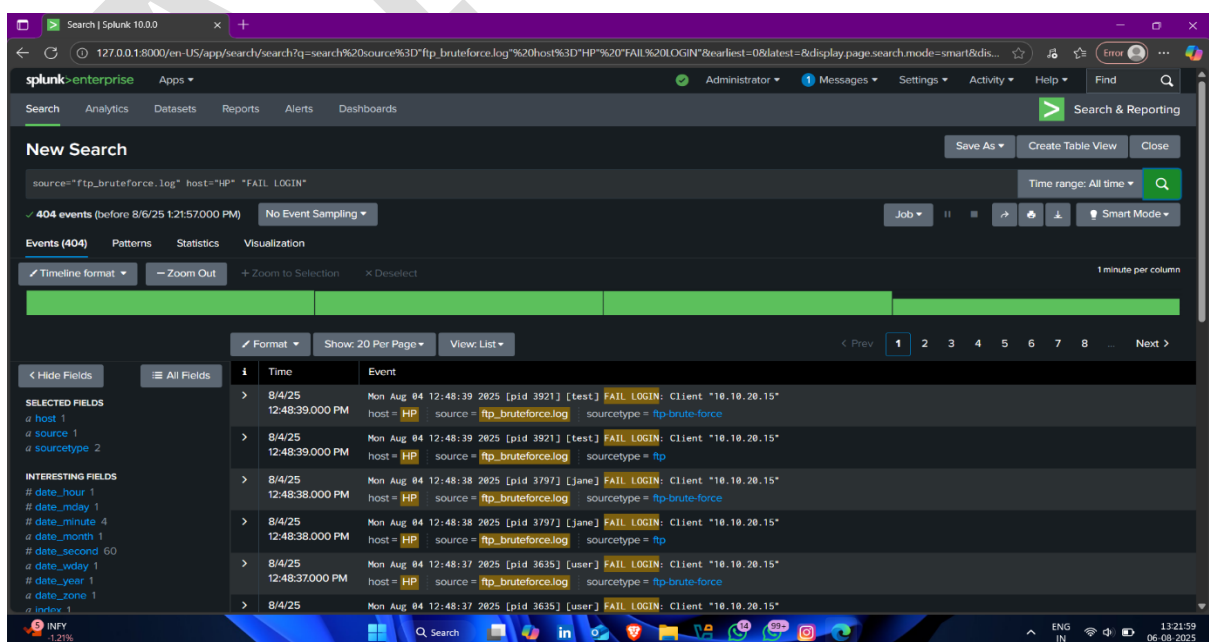
[Back](#) [Submit](#)

- Now click on **Start Searching**



Result - The logs show "**FAIL LOGIN**" events, meaning login attempts failed.

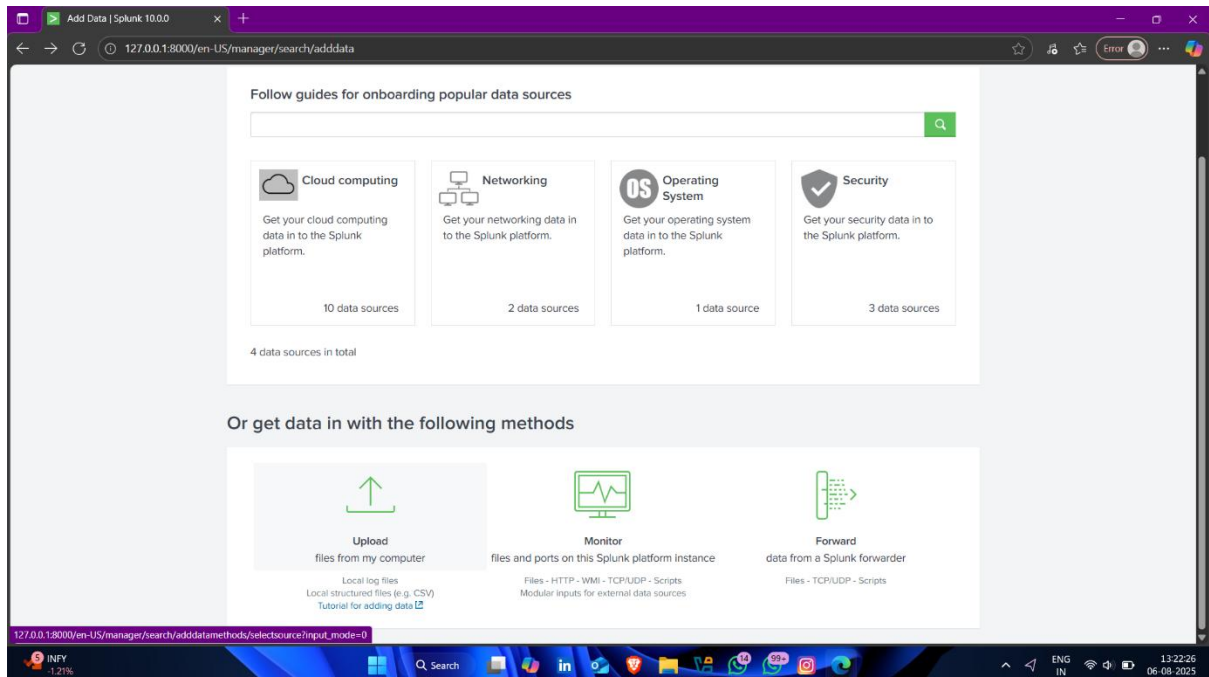
- The **client IP** 10.10.20.15 is trying to log in repeatedly.
- Different **usernames** were tried: test, jane, user.
- This pattern of repeated failed logins is typical of a **brute force attack**, where an attacker tries many username/password combinations to guess the correct one.



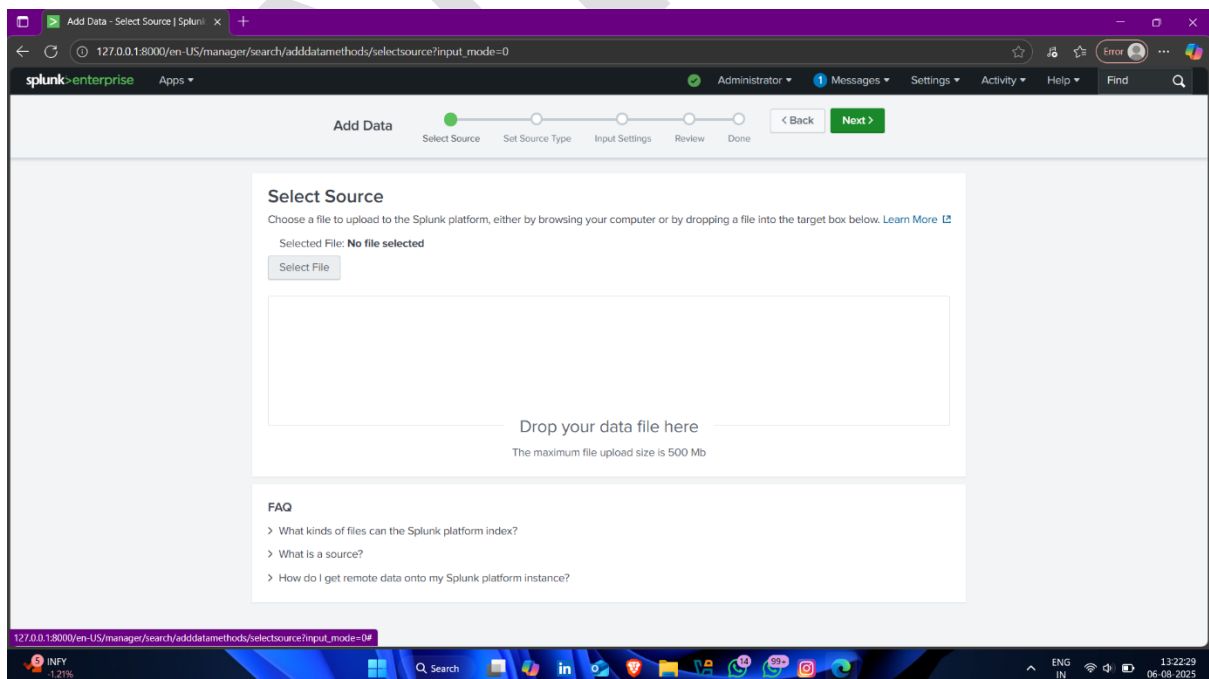
Sql Injection Attack Log Analysis Using Splunk

How to do it -:

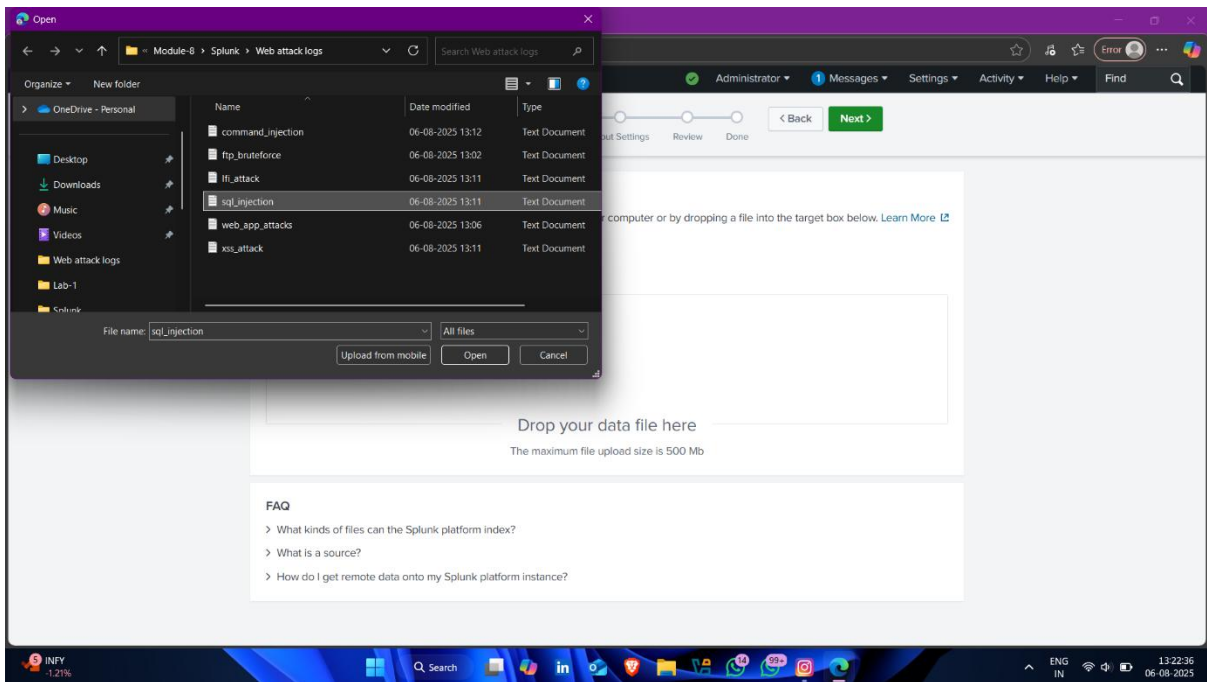
- Click on **Upload** Option



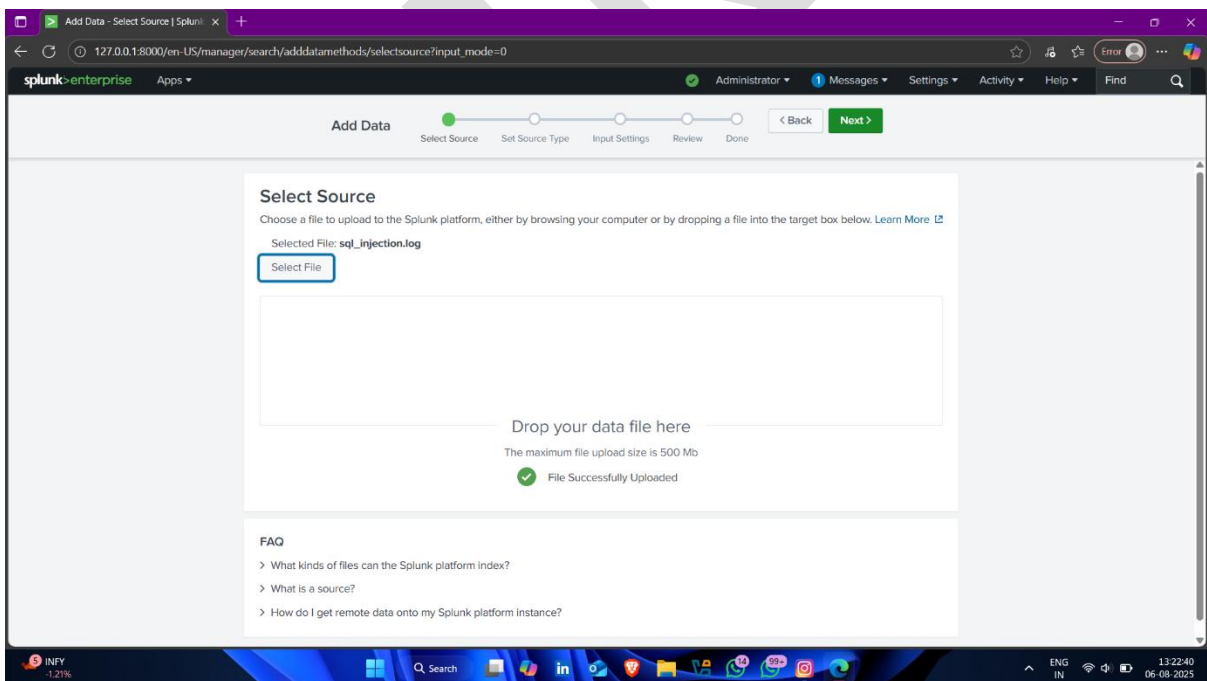
- Click on **Select file**



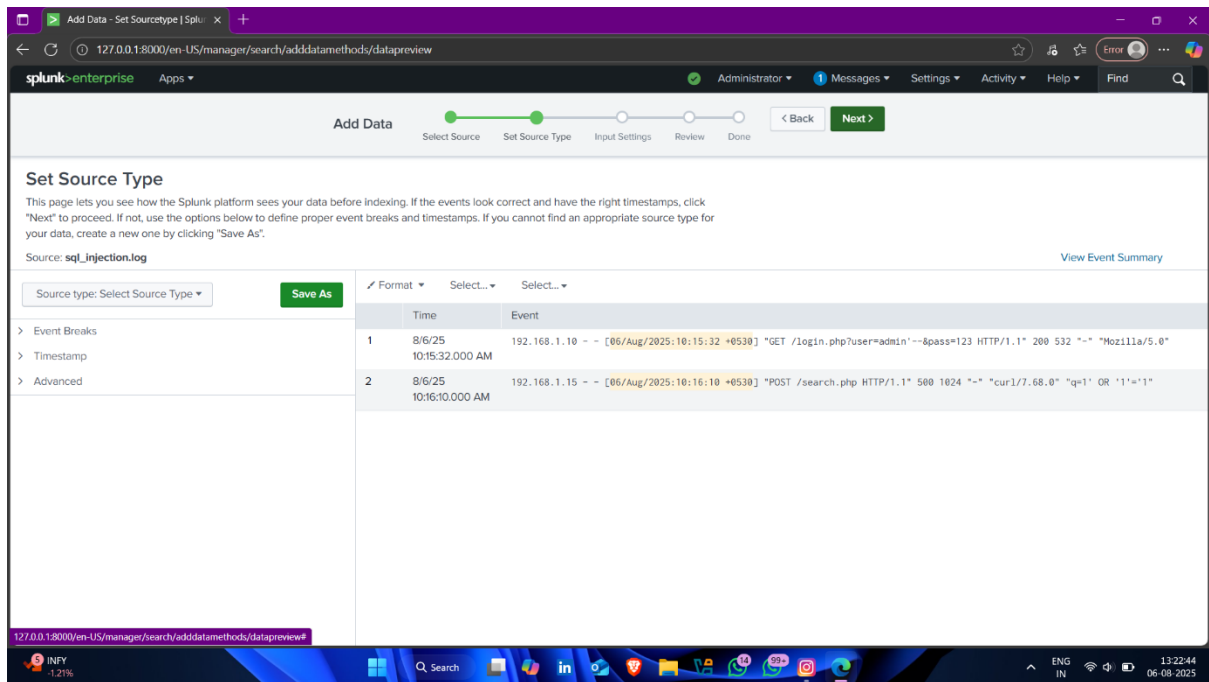
- Select **Sql-Injection** Log file and then click on **open**



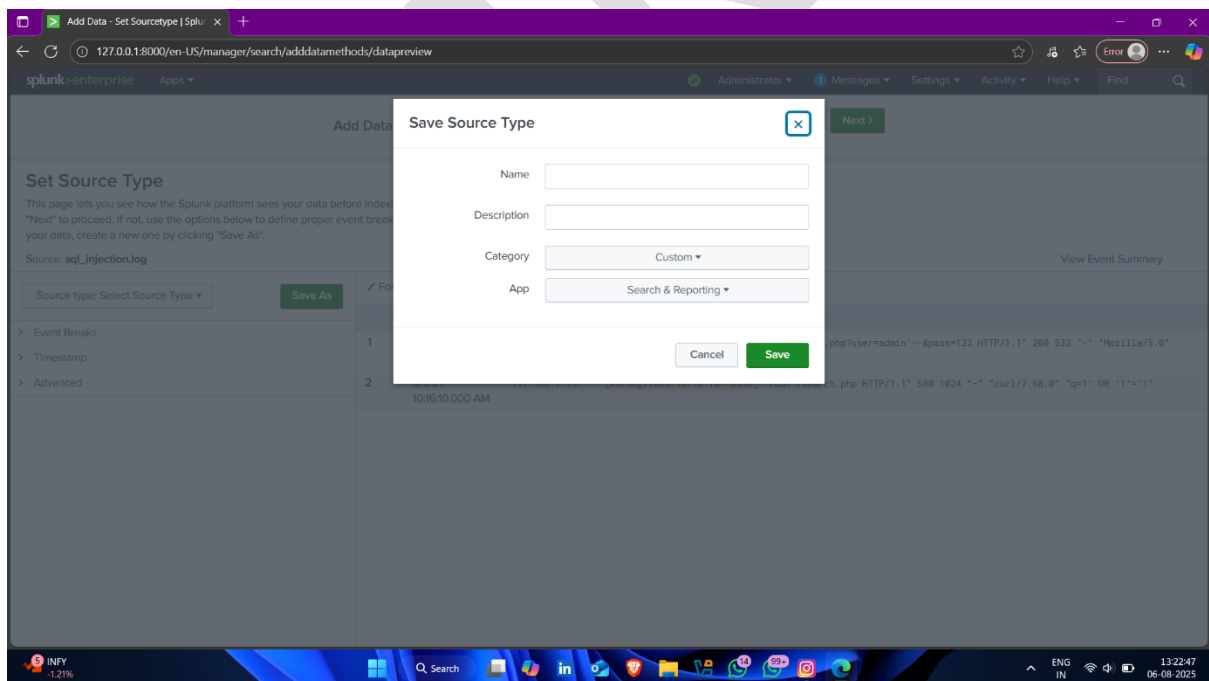
- Log Uploaded , Now click on **next**



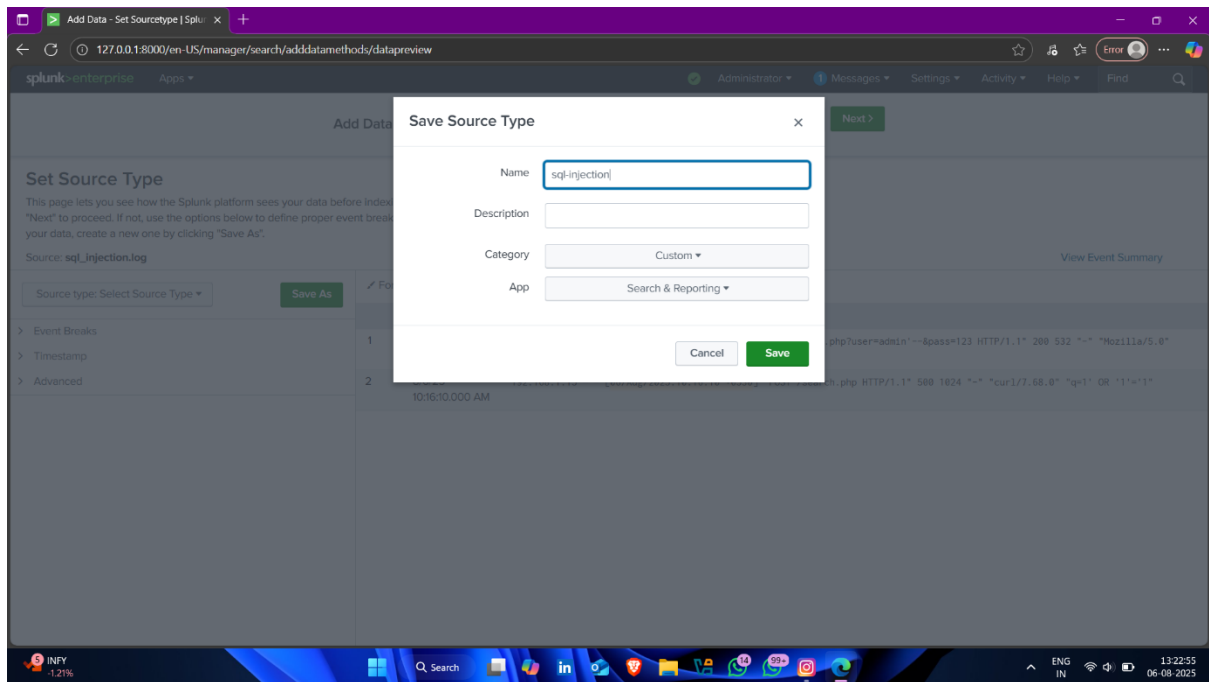
- Click on Next



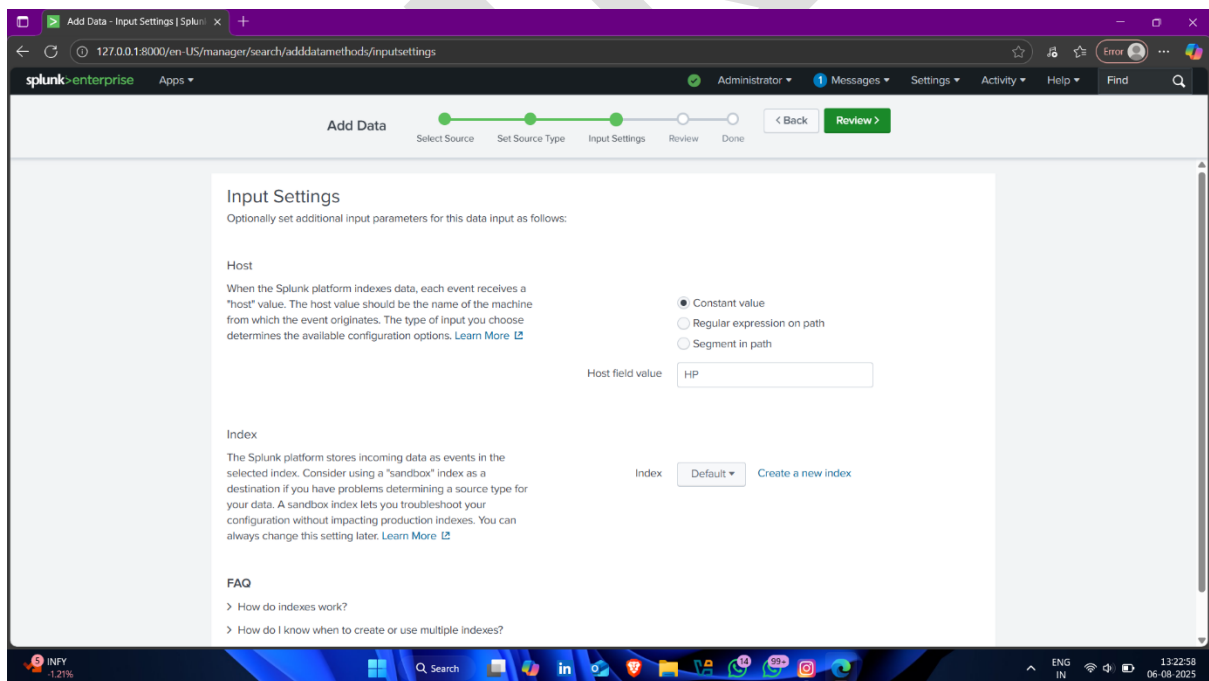
- Enter Source file name



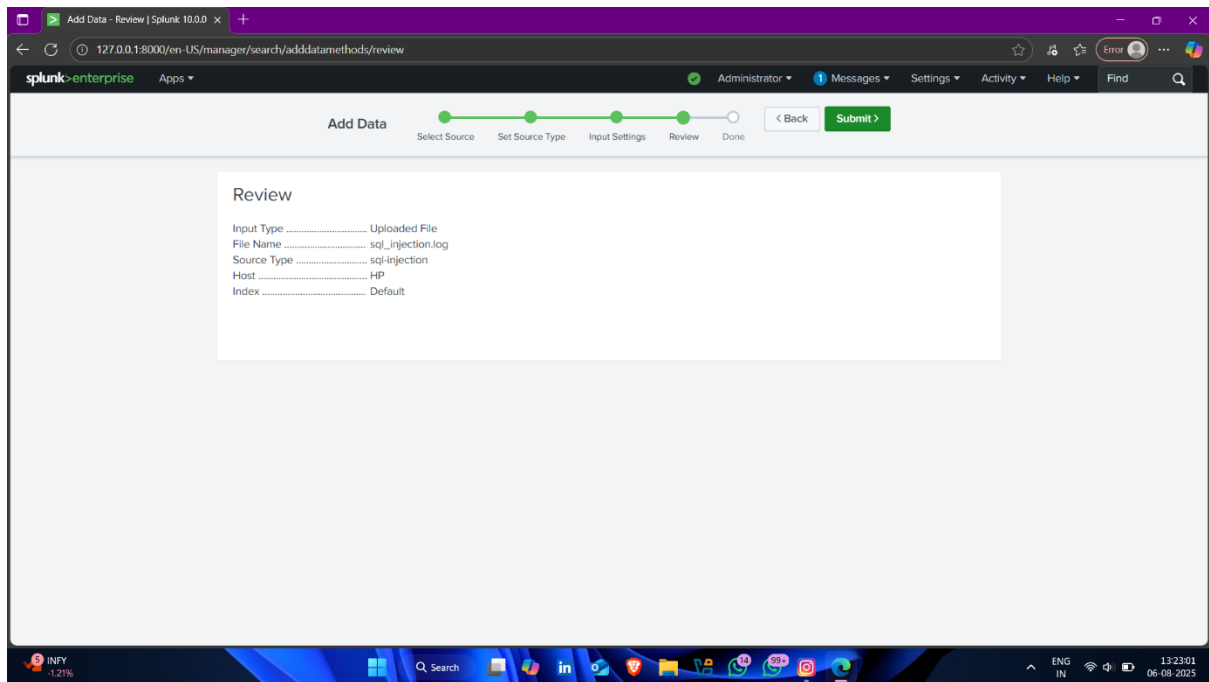
- Click on **Save**



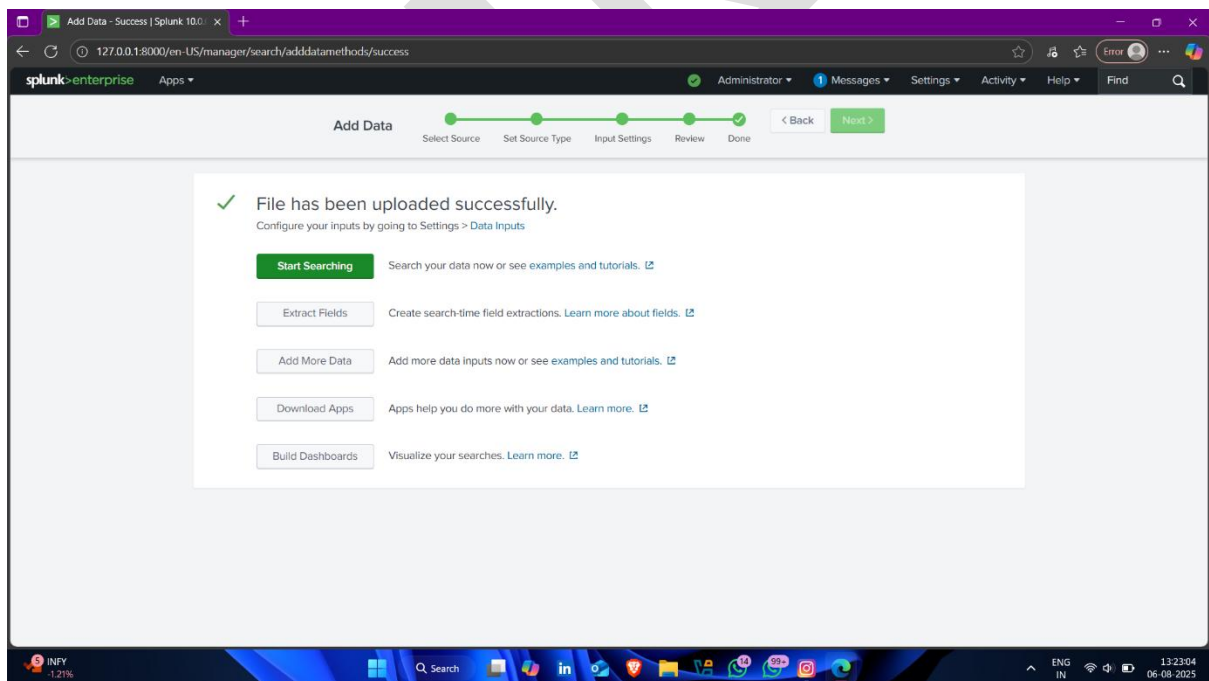
- Click on **Review**



- Click on **Submit**



- Now click on **start Searching**

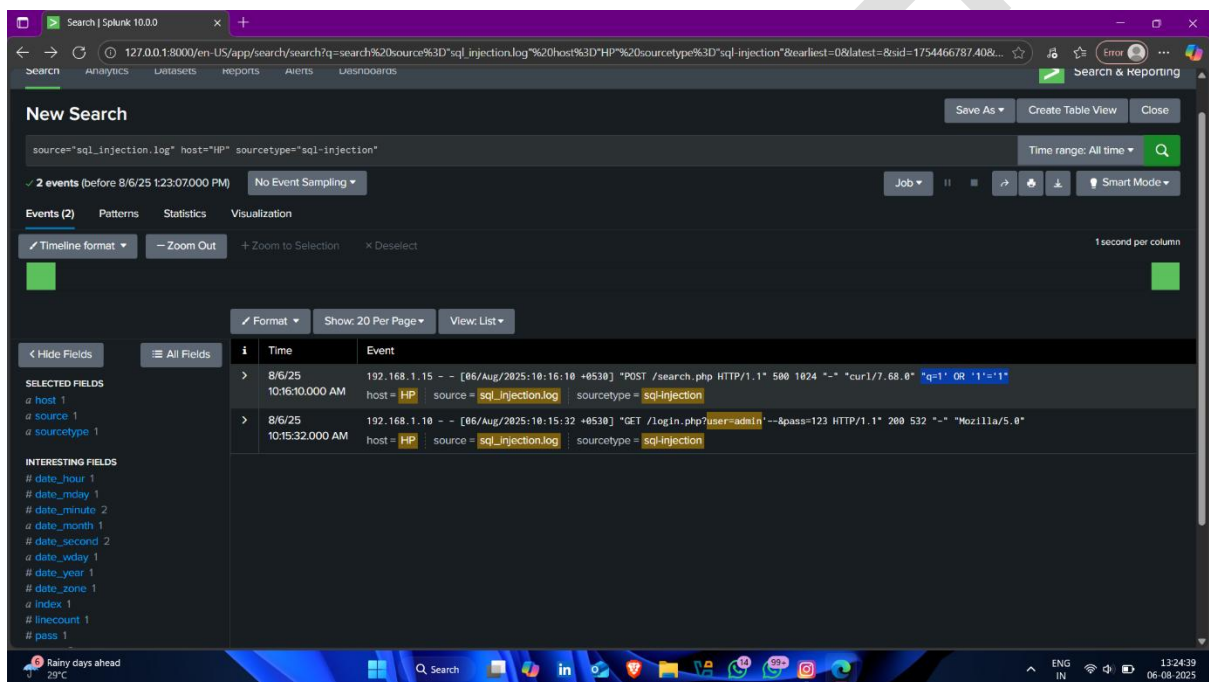


Result -: Tried to hack search.php by typing **1' OR '1'='1** in the search box.

- This trick can force the database to show all data.
- The attack failed and caused a server error (500).

Tried to log in to login.php as admin by adding ' -- in the username.

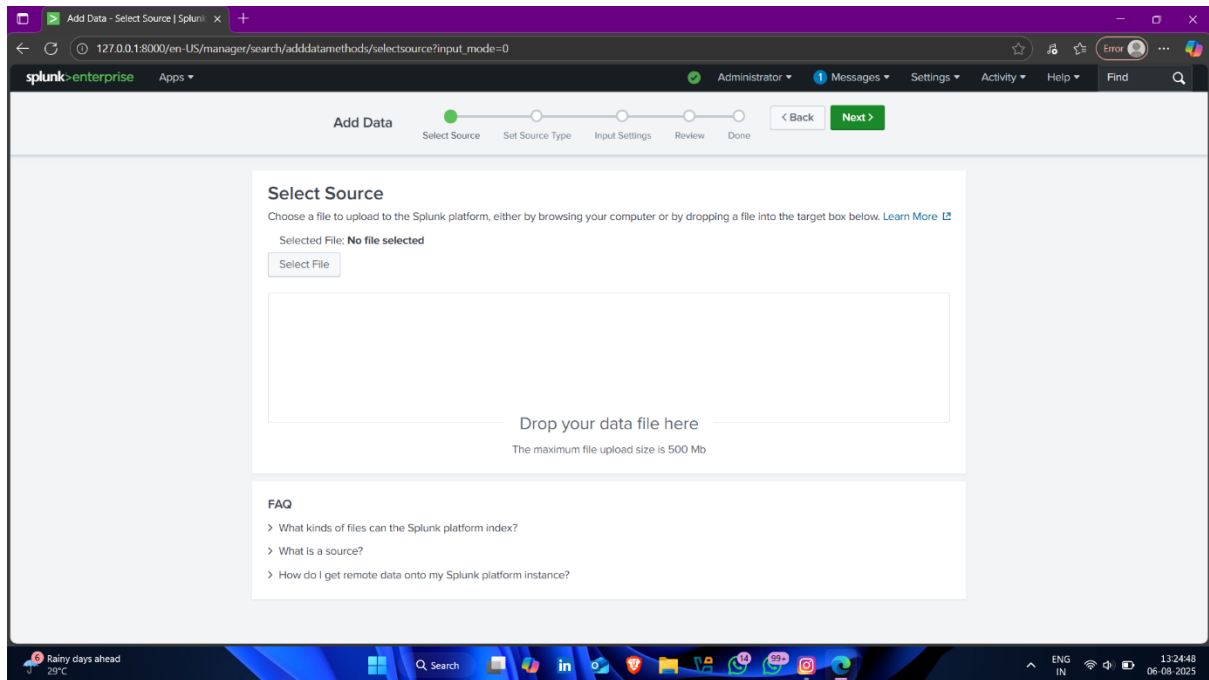
- This can bypass the password check.
- The page loaded successfully (200), so it might have worked.



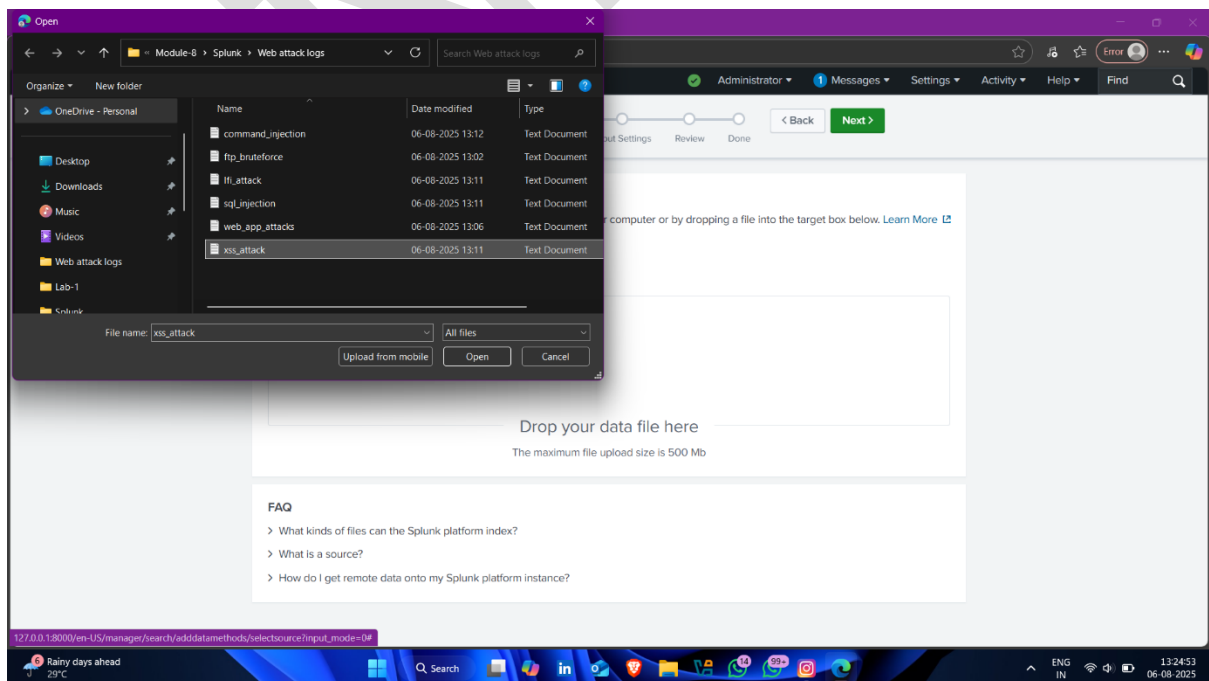
Cross Site Scripting (XSS) Attack Log Analysis Using Splunk

How to do it -:

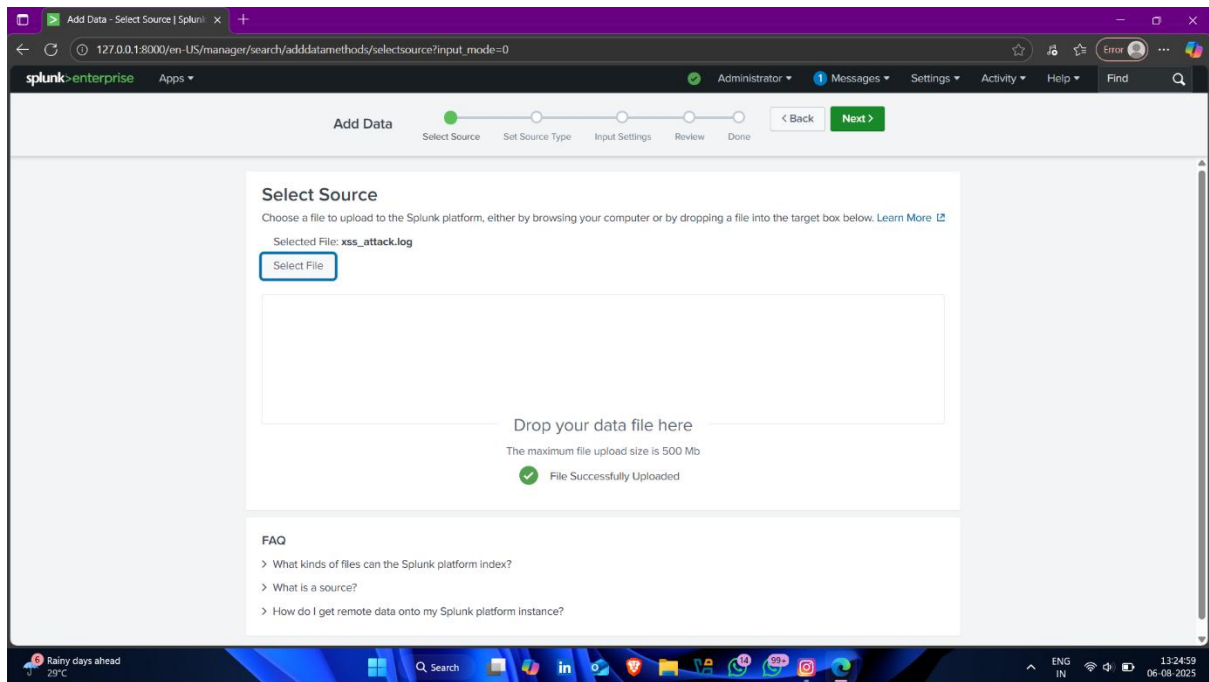
- Click on **Select File**



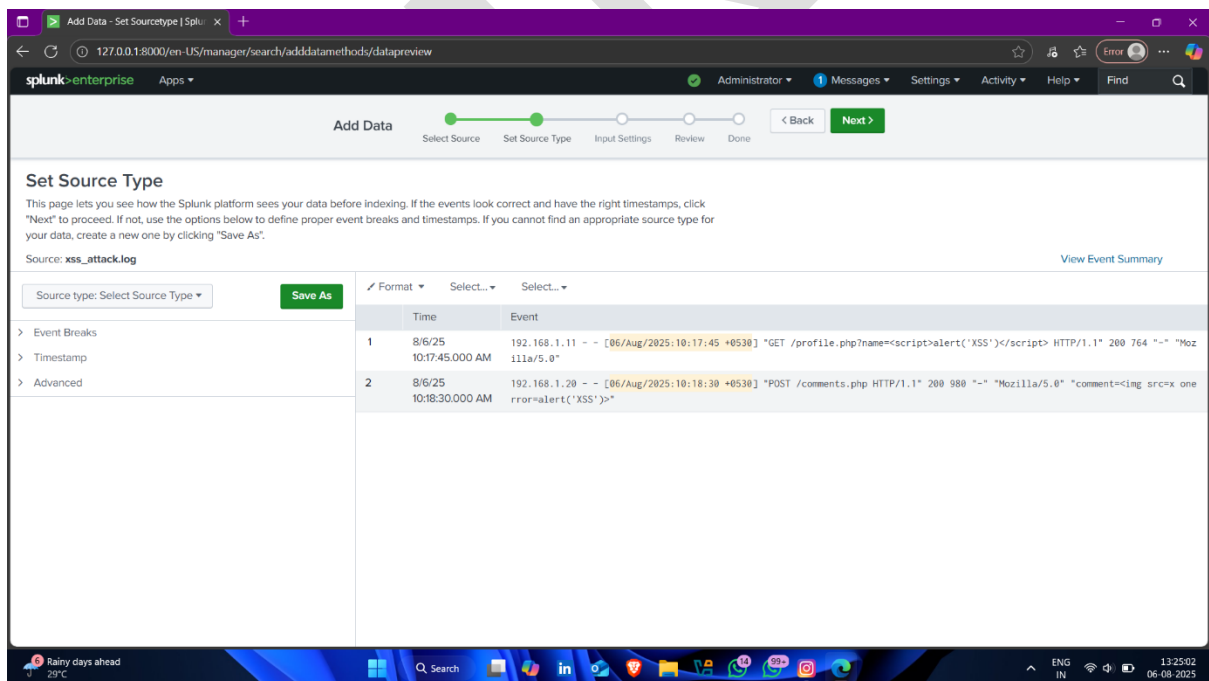
- Click on **Xss Log File** and then click on **Open**



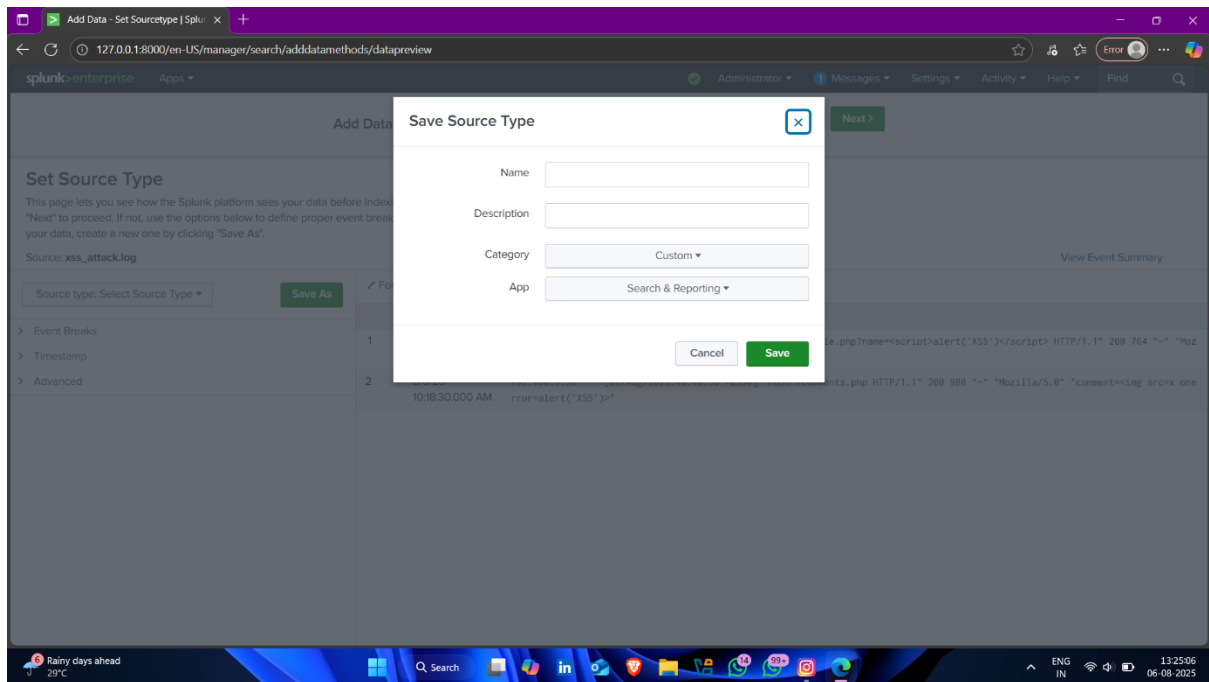
- Log file Uploaded , click on **Next**



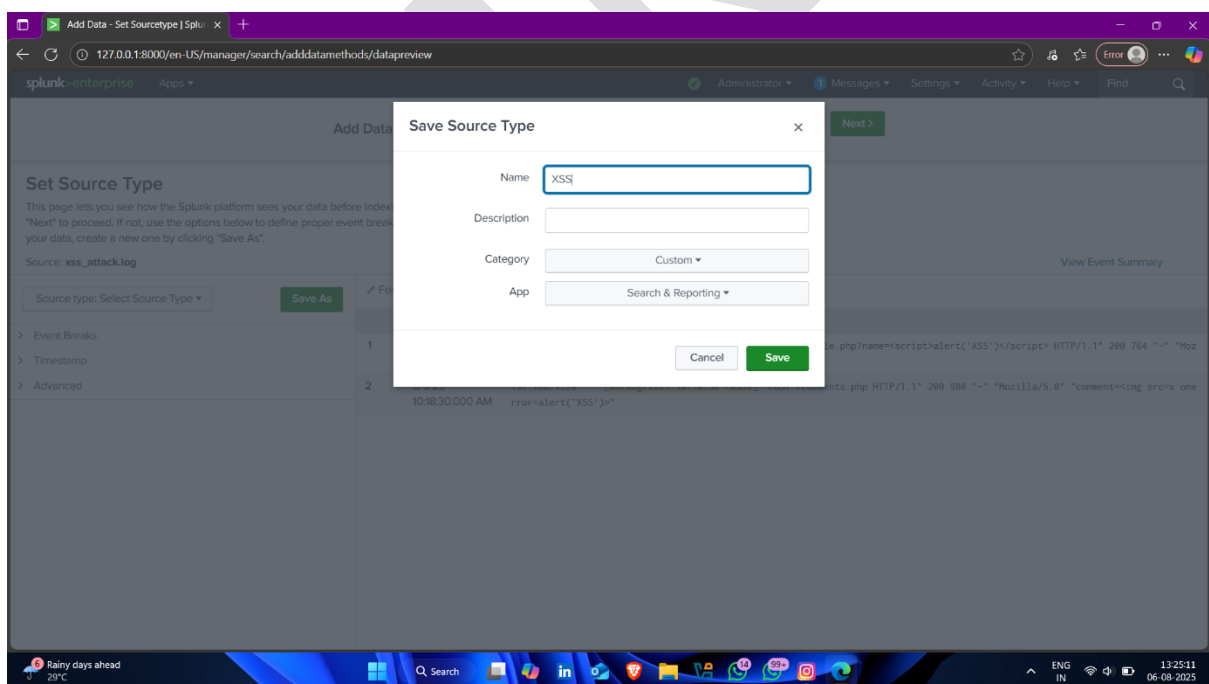
- Click on **Next**



- Enter source file name



- Click on **save**



- click on **Review**

Add Data | Select Source | Set Source Type | **Input Settings** | Review | Done

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[Back](#) [Review](#)

- click on **Submit**

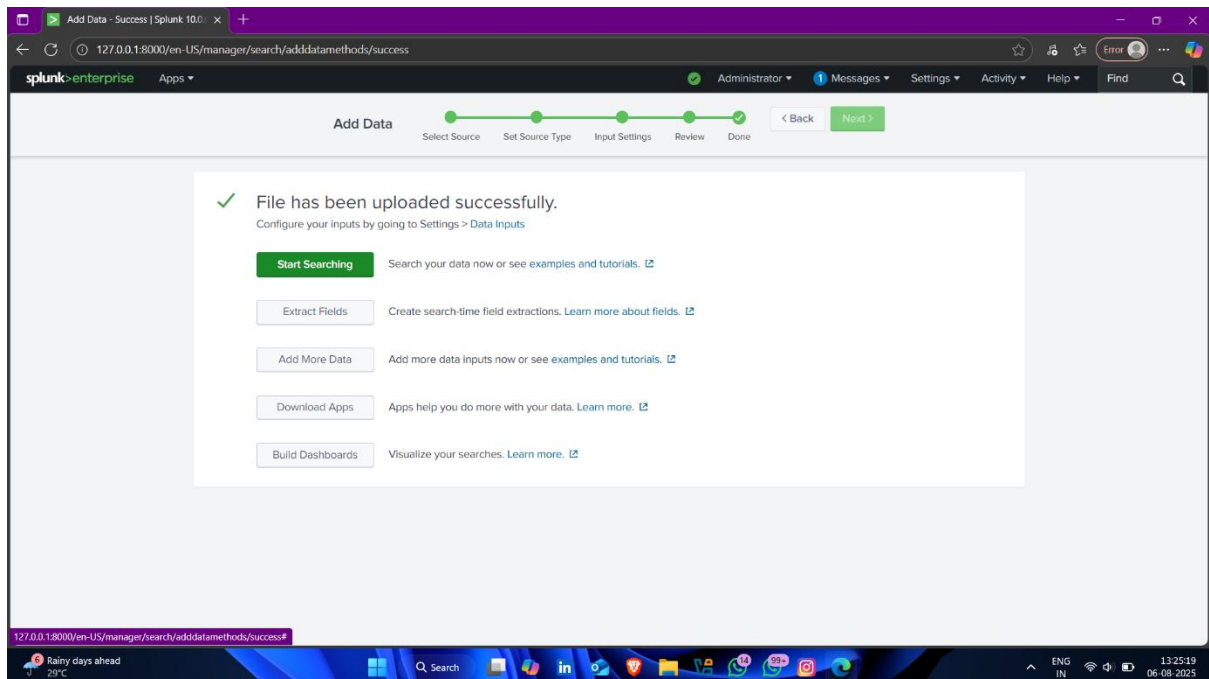
Add Data | Select Source | Set Source Type | Input Settings | **Review** | Done

Review

Input Type Uploaded File
File Name xss_attack.log
Source Type XSS
Host HP
Index Default

[Back](#) [Submit](#)

- Now click on **Start Searching**



- Result - ☐ osted a comment on comments.php with malicious code:

```
<img src=x onerror=alert('XSS')>
```

- This would pop up an alert if someone views the comment.
- Server responded **200** (success), so it was stored.
- Visited profile.php and inserted malicious script in the name field:

```
<script>alert('XSS')</script>
```

- If someone views that profile, a popup will appear.
- Server responded **200**, so the attack likely worked.

Splunk Enterprise interface showing a search for XSS attacks. The search query is `source="xss_attack.log" host="HP" sourcetype="XSS"`. The results show two events from 8/6/25.

Search Query: `source="xss_attack.log" host="HP" sourcetype="XSS"`

Results:

Time	Event
8/6/25 10:18:30.000 AM	192.168.1.20 - - [06/Aug/2025:10:18:30 +0530] "POST /comments.php HTTP/1.1" 200 980 "-" "Mozilla/5.0" "comment=" host=HP source=xss_attack.log sourcetype=XSS
8/6/25 10:17:45.000 AM	192.168.1.11 - - [06/Aug/2025:10:17:45 +0530] "GET /profile.php?name=<script>alert('XSS')</script> HTTP/1.1" 200 764 "-" "Mozilla/5.0" host=HP source=xss_attack.log sourcetype=XSS

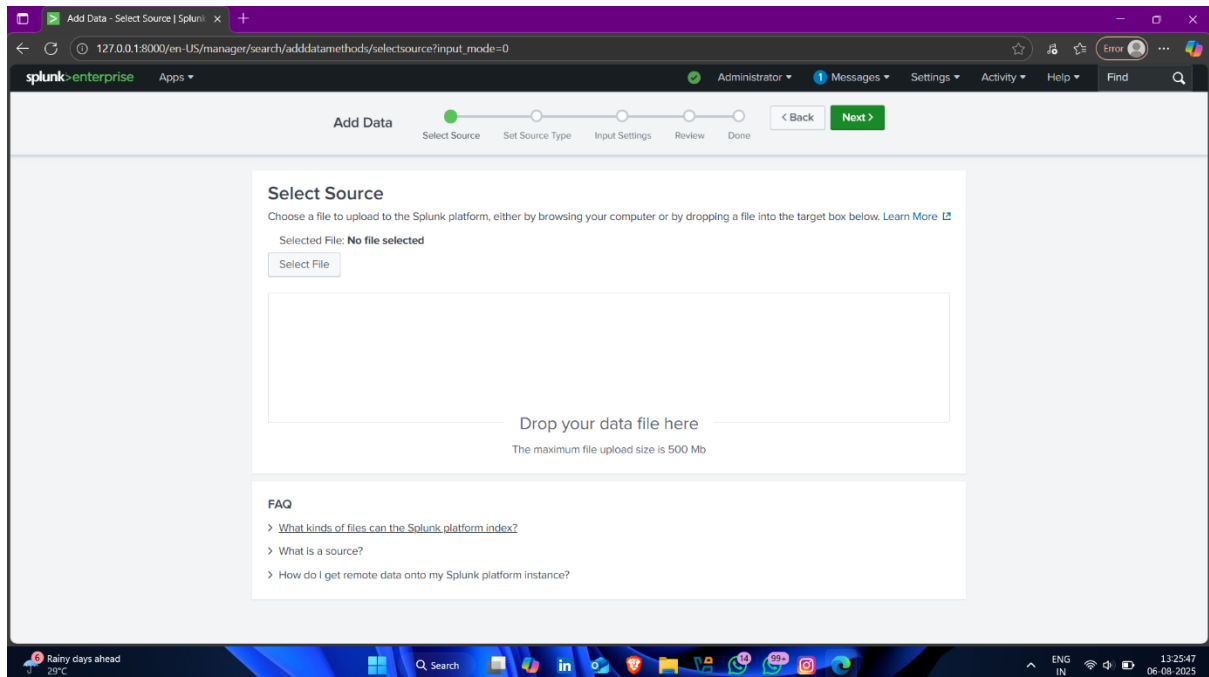
Selected Fields: host, source, sourcetype

Interesting Fields: comment, date_hour, date_mday, date_minute, date_month, date_second, date_wday, date_year, date_zone

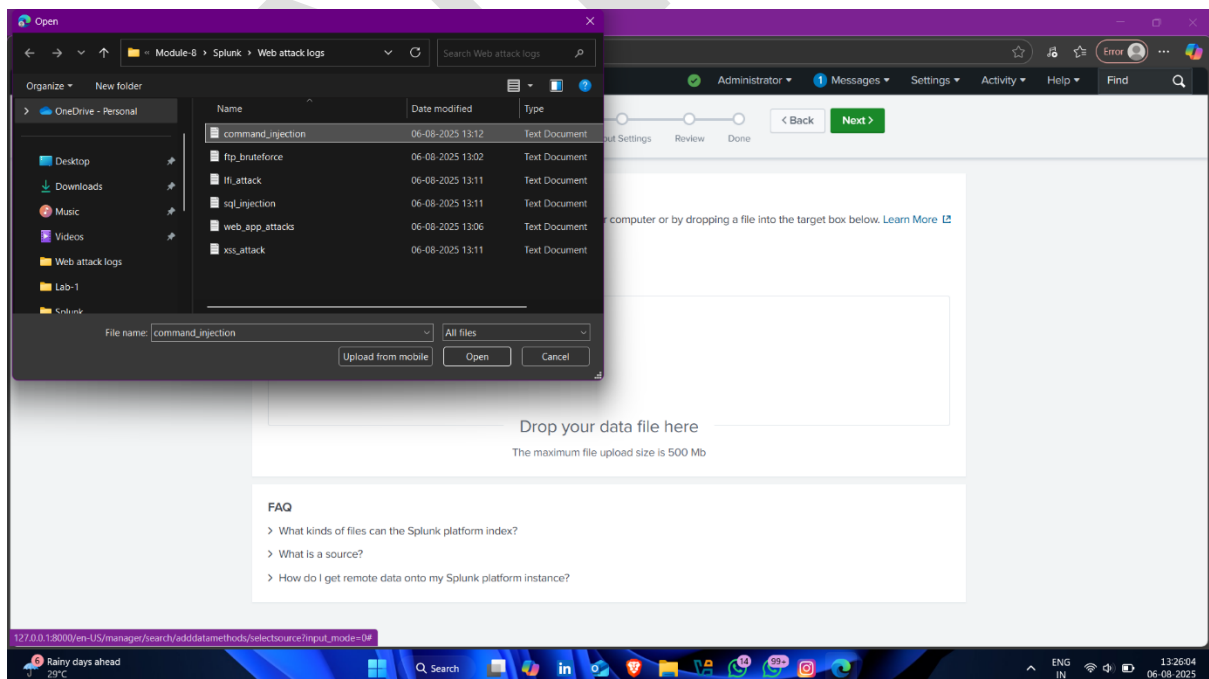
Command Injection Attack Log Analysis Using Splunk

How to do it -:

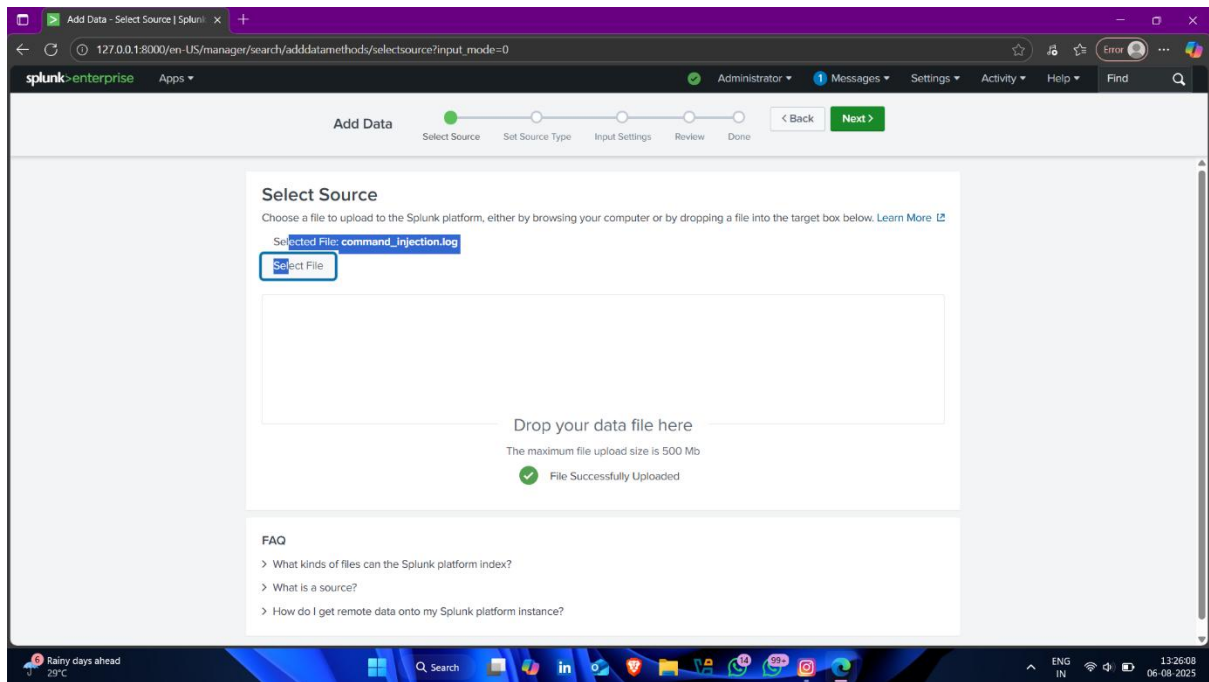
- Click on **Select File**



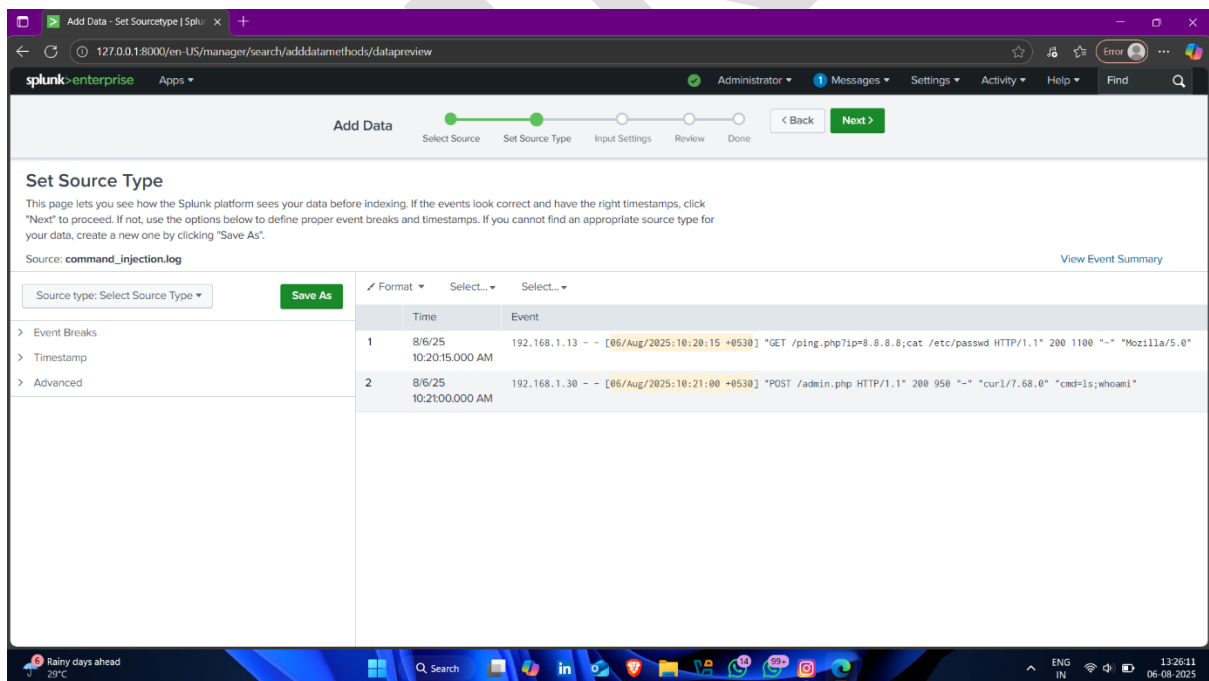
- Select **Command injection log file** and then click on **Open**



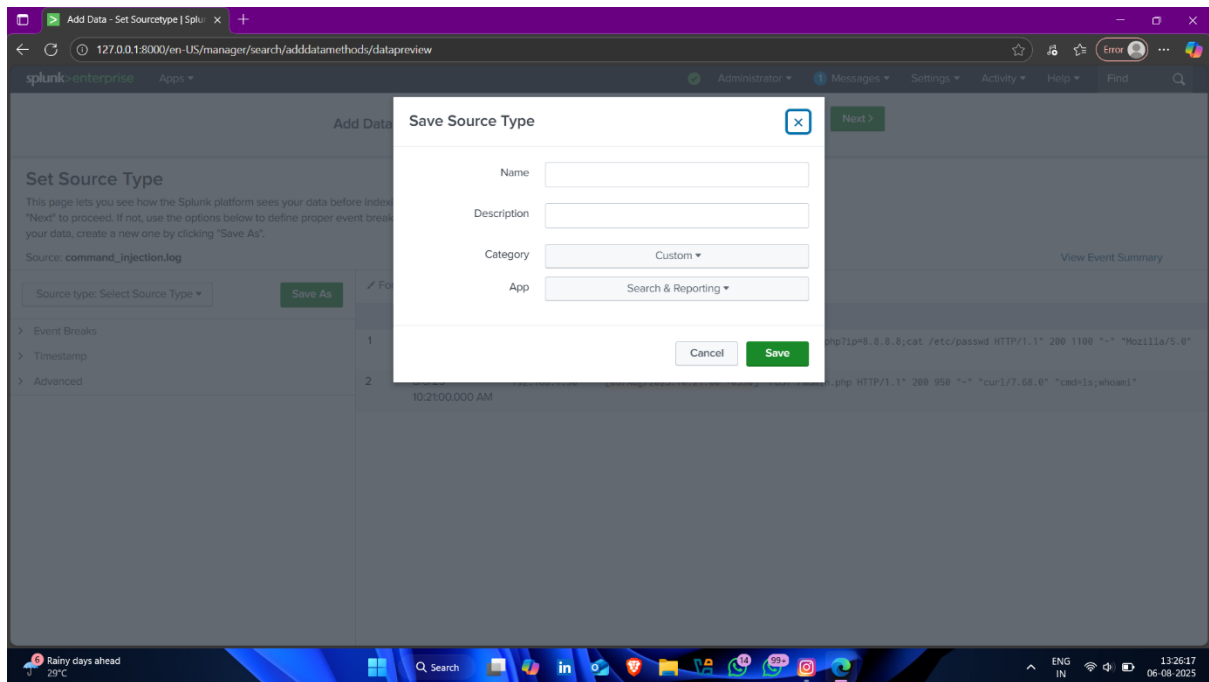
- Log File Uploaded , click on **next**



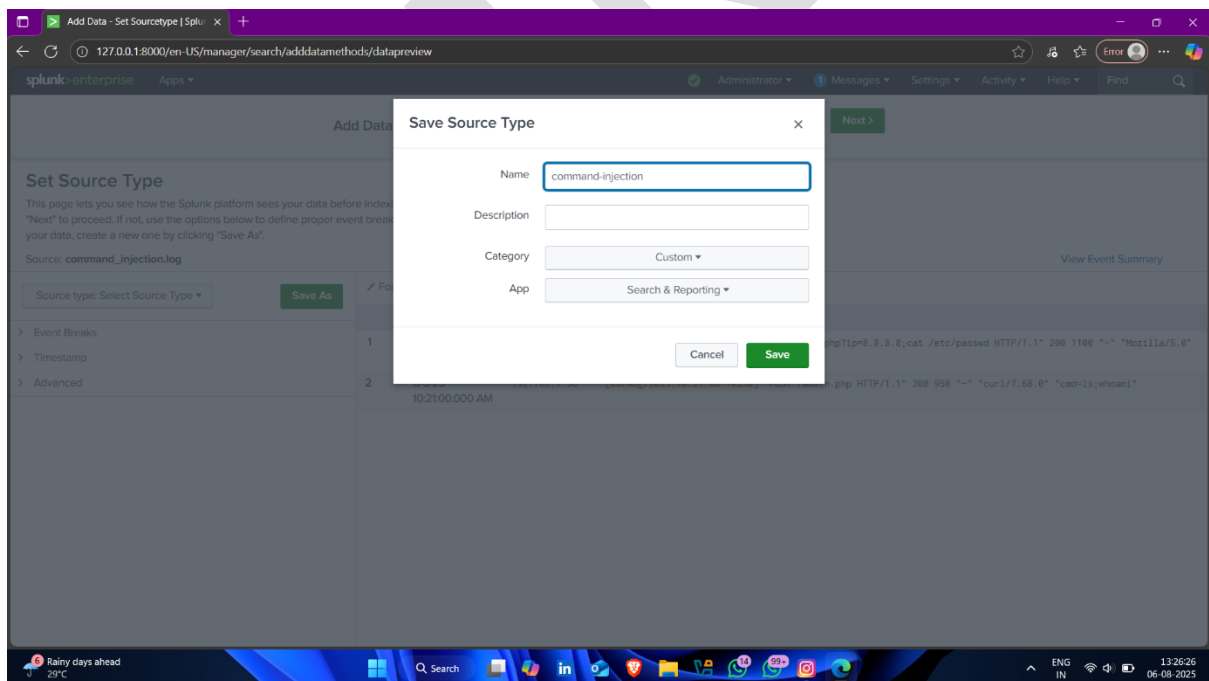
- Click on **Next**



- Enter source file name



- Click on **Save**



- Click on **Review**

Add Data | Select Source | Set Source Type | **Input Settings** | Review | Done | < Back | **Review >**

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Host field value:

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: [Create a new index](#)

FAQ

- > How do indexes work?
- > How do I know when to create or use multiple indexes?

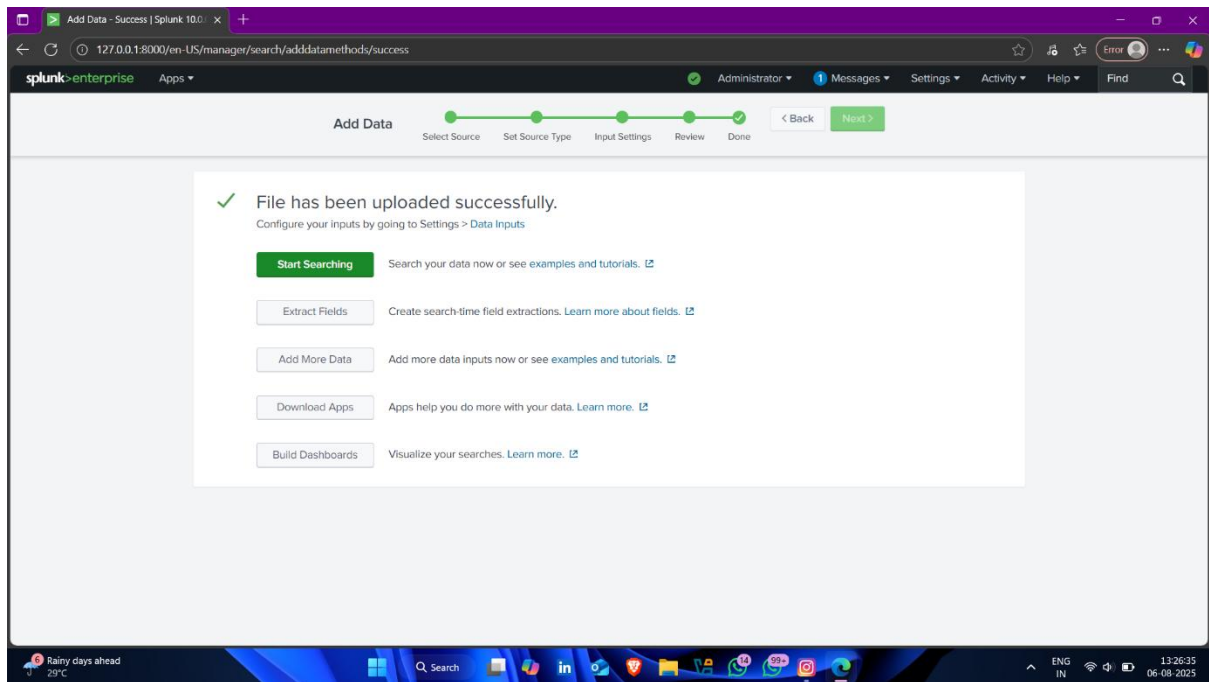
- Click on **Submit**

Add Data | Select Source | Set Source Type | Input Settings | **Review** | Done | < Back | **Submit >**

Review

Input Type Uploaded File
File Name command_injection.log
Source Type command-injection
Host HP
Index Default

- Click on **Start Searching**



Result -: Sent a request to admin.php with malicious commands:

cmd=ls;whoami

- This tries to list files (ls) and show the current user (whoami).
- Server responded **200** (success), so the command likely ran.

Sent a request to **ping.php** with extra command:

8.8.8.8;cat /etc/passwd

- This tries to display the system's /etc/passwd file (which contains user account info).
- Server responded **200**, so the command likely executed.

Search | Splunk 10.0.0

127.0.0.1:8000/en-US/app/search/search?q=search%20source%3D%20command_injection.log%20host%3D%20HP%20sourcetype%3D%20command-injection&earliest=0&latest=&sid=1754466...

splunk-enterprise Apps

Administrator Messages Settings Activity Help Find

Search Analytics Datasets Reports Alerts Dashboards

Search & Reporting

New Search

Save As Create Table View Close

source="command_injection.log" host="HP" sourcetype="command-injection" Time range: All time

2 events (before 8/6/25 1:26:38.000 PM) No Event Sampling Job

Events (2) Patterns Statistics Visualization

Timeline format Zoom Out + Zoom to Selection X Deselect 1 second per column

Format Show: 20 Per Page View: List

i	Time	Event
>	8/6/25 10:21:00.000 AM	192.168.1.38 - - [06/Aug/2025:10:21:00 +0530] "POST /admin.php HTTP/1.1" 200 950 "-" "curl/7.68.0" "cmd=ls;whoami"
>	8/6/25 10:20:15.000 AM	192.168.1.13 - - [06/Aug/2025:10:20:15 +0530] "GET /ping.php?ip=8.8.8.8;cat /etc/passwd HTTP/1.1" 200 1100 "-" "Mozilla/5.0"

SELECTED FIELDS

- a host 1
- a source 1
- a sourcetype 1

INTERESTING FIELDS

- a cmd 1
- # date_hour 1
- # date_mday 1
- # date_minute 2
- # date_month 1
- # date_second 2
- a date_wday 1
- # date_year 1
- # date_zone 1

NIFTY -0.15%

Q Search

ENG IN 13:26:55 06-08-2025

Lab-3 Identify And Investigate Various Network Attacks Using Wireshark

Wireshark – Complete Guide

1. Introduction

Wireshark is a **free and open-source network protocol analyzer** used to capture, inspect, and analyze network traffic in real time. It's widely used for:

- Network troubleshooting
 - Performance analysis
 - Cybersecurity investigations
 - Digital forensics
 - Protocol development and debugging
-

2. Key Features

- **Live packet capture** from wired, wireless, USB, and virtual interfaces.
 - **Packet filtering** using display filters.
 - **Protocol support:** 3000+ protocols including TCP, UDP, HTTP, FTP, DNS, SSL/TLS, SMB, etc.
 - **Color coding** for quick analysis.
 - **Packet reassembly** (e.g., reconstruct HTTP files, VoIP calls).
 - **Export options** to multiple formats (pcap, CSV, JSON, etc.).
 - **Decryption support** for SSL/TLS, WPA/WPA2 (if keys are available).
-

3. Wireshark Architecture

Wireshark uses two main components:

1. **Packet Capture Library** – *libpcap* on Linux/macOS, *WinPcap/Npcap* on Windows.

2. **Wireshark GUI** – for packet visualization and analysis.

5. **Capturing Traffic**

- Choose the interface (e.g., eth0, wlan0).
- Click **Start Capturing**.
- Stop capture when enough data is collected.

Tip: Use capture filters to limit data collected (e.g., only HTTP traffic).

6. **Capture Filters vs Display Filters**

Capture Filters (before capture, BPF syntax):

host 192.168.1.10 # Capture only traffic to/from host

port 80 # Capture only port 80 traffic

tcp # Capture only TCP traffic

Display Filters (after capture, Wireshark syntax):

ip.src == 192.168.1.5

http

tcp.port == 443

dns.qry.name contains "google.com"

7. **Analyzing Packets**

- **Packet List Pane** – all captured packets.
- **Packet Details Pane** – protocol breakdown.
- **Packet Bytes Pane** – raw hexadecimal and ASCII.

Example: Inspecting an HTTP GET request:

- Filter: http.request
 - Look at Host, User-Agent, GET path.
-

8. Common Analysis Scenarios

1. Detecting FTP Brute Force

`ftp && tcp.analysis.retransmission == 0`

Look for multiple 530 Login incorrect responses from the same IP.

2. Finding Malware C2 Traffic

- Look for **unusual DNS requests**:

`dns && !(dns.qry.name contains "knownsite.com")`

- Suspicious HTTP traffic with strange User-Agent strings.

3. Extracting Files

- Filter: http
- Follow → HTTP Stream → Save file.
- Or use: **File** → **Export Objects** → **HTTP/FTP/DICOM/SMB**.

4. VoIP Call Analysis

- Filter: rtp
- Telephony → VoIP Calls → Play Streams.

9. Security & Forensics Use

- **Incident Response**: Identify compromised hosts, attack vectors, and exfiltrated data.
- **Law Enforcement**: Collect packet captures for evidence.
- **Threat Hunting**: Spot anomalies in traffic patterns.
- **Post-Breach Analysis**: Trace attacker movement.

10. Performance Troubleshooting

- **High Latency**: Look for high TCP round-trip times (`tcp.analysis.ack_rtt`).
- **Packet Loss**: Check `tcp.analysis.lost_segment`.
- **Retransmissions**: `tcp.analysis.retransmission` filter.

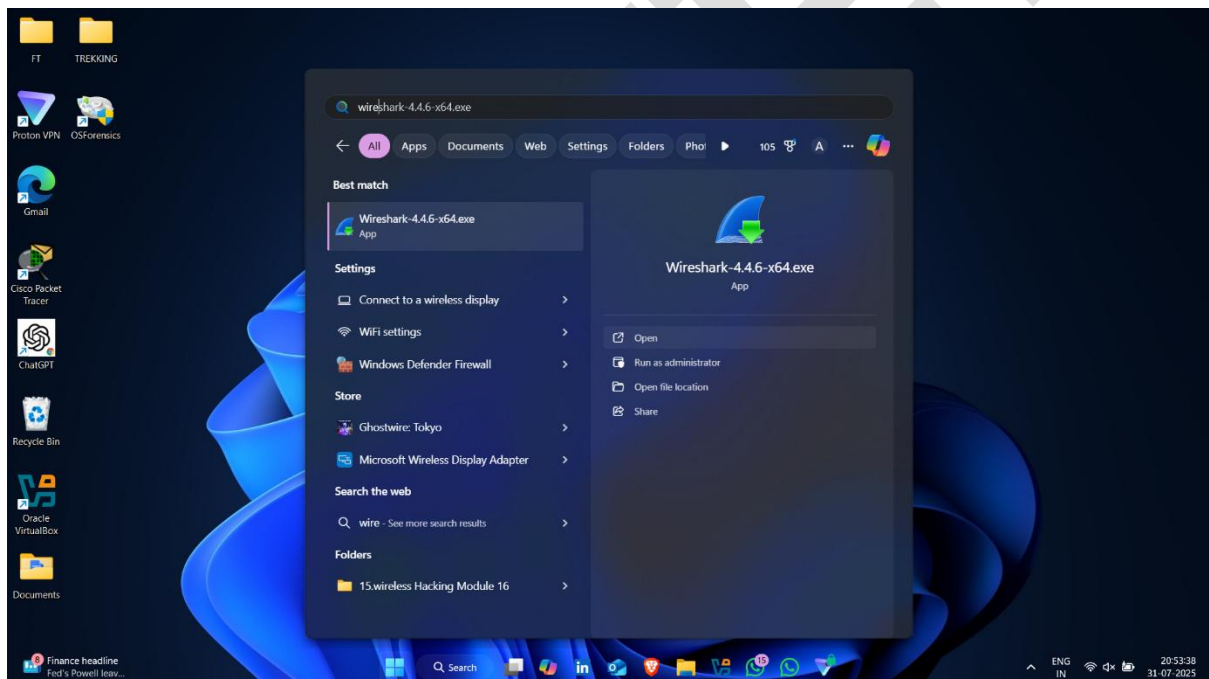
11. Exporting Data

Wireshark can export:

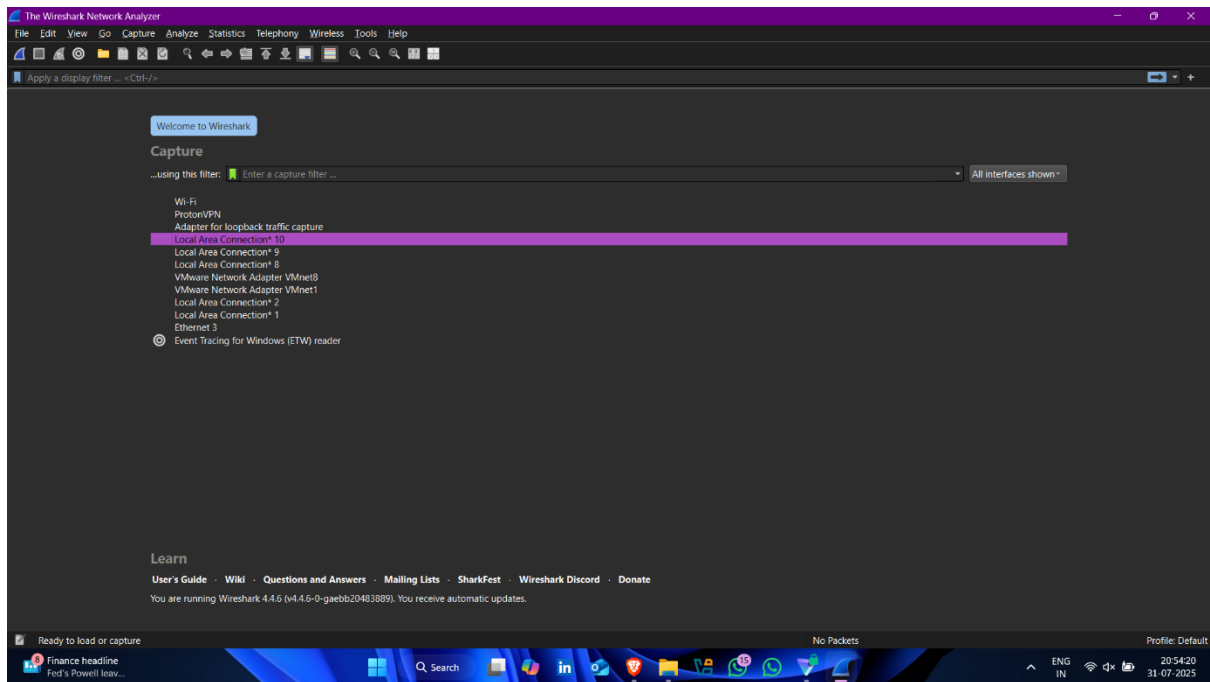
- Specific packets (File → Export Specified Packets).
 - Protocol objects (e.g., files transferred).
 - Statistics (CSV/JSON for reporting).
-

How to do it -:

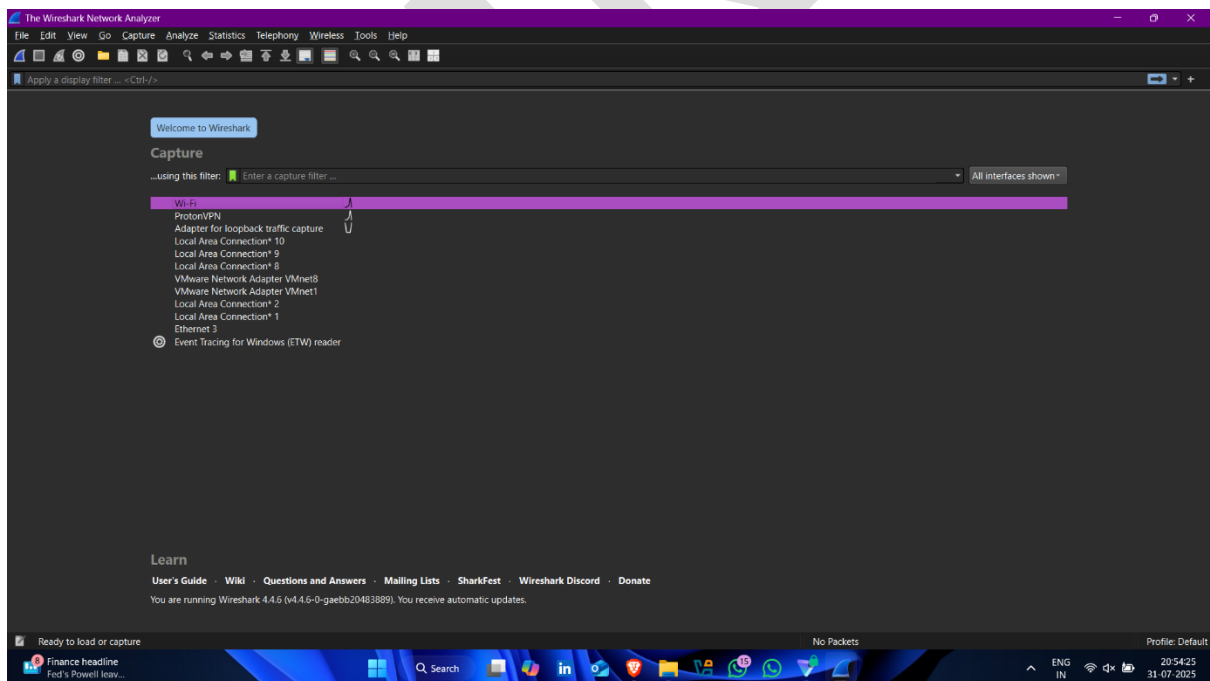
- Start Wireshark



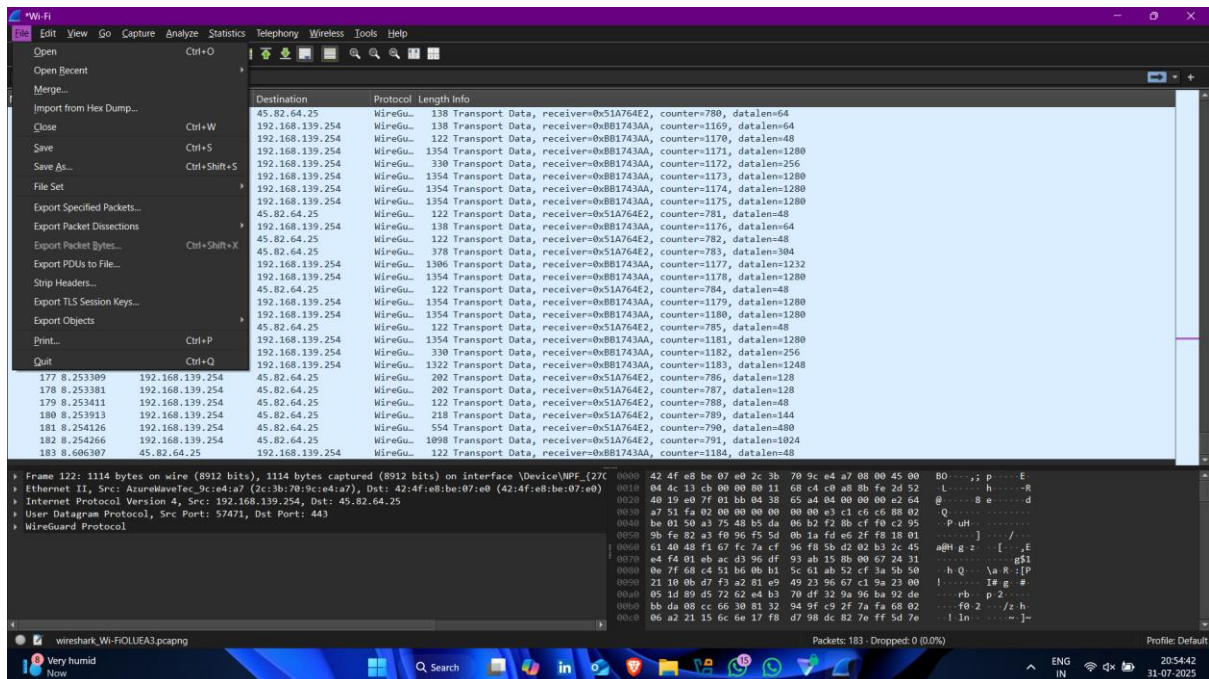
- **Select Network Interface**



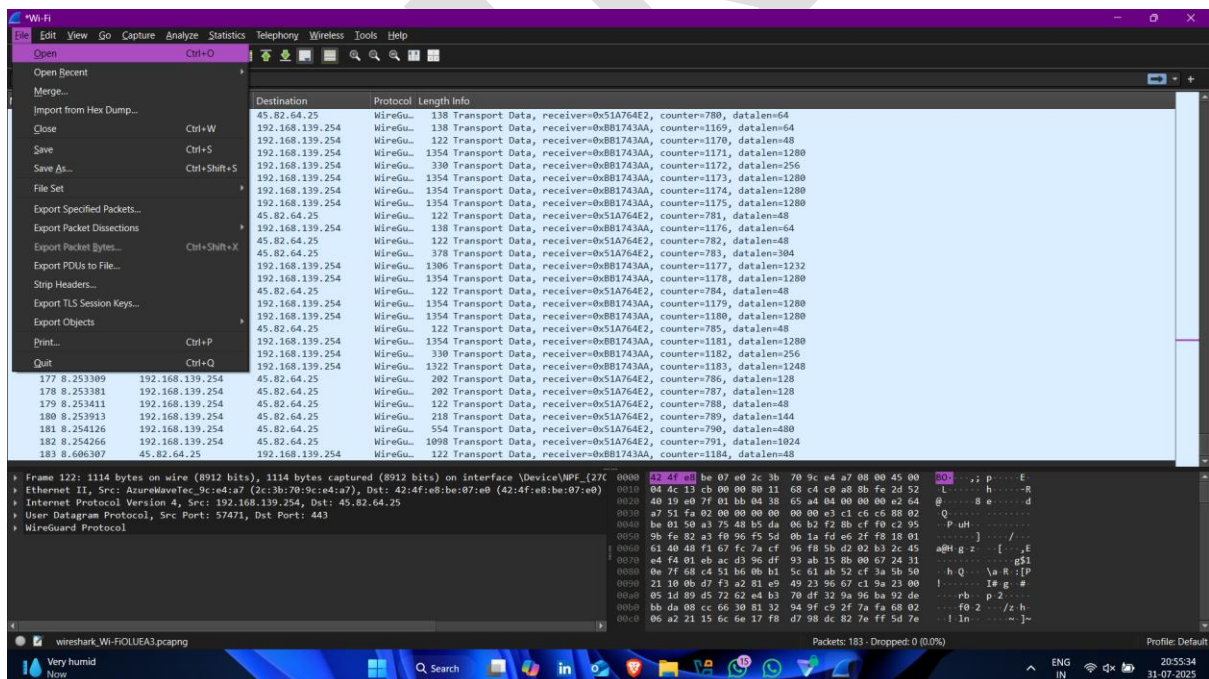
- **Click on Network Interface**



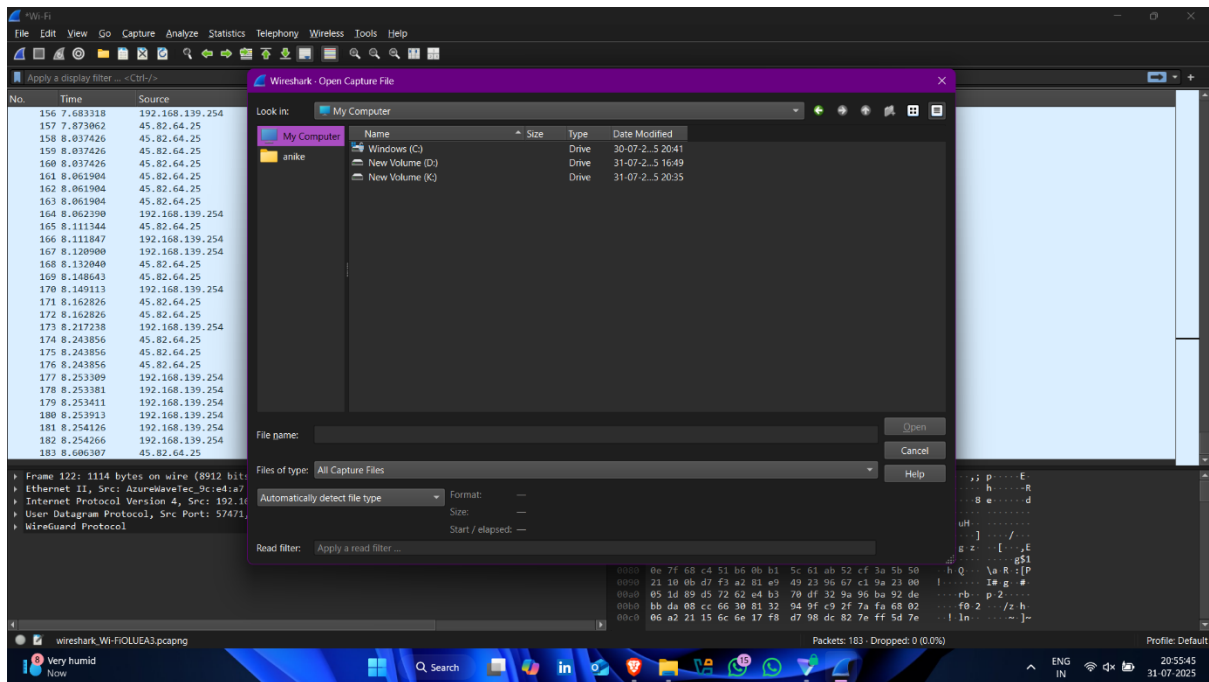
- Click on **File** Section



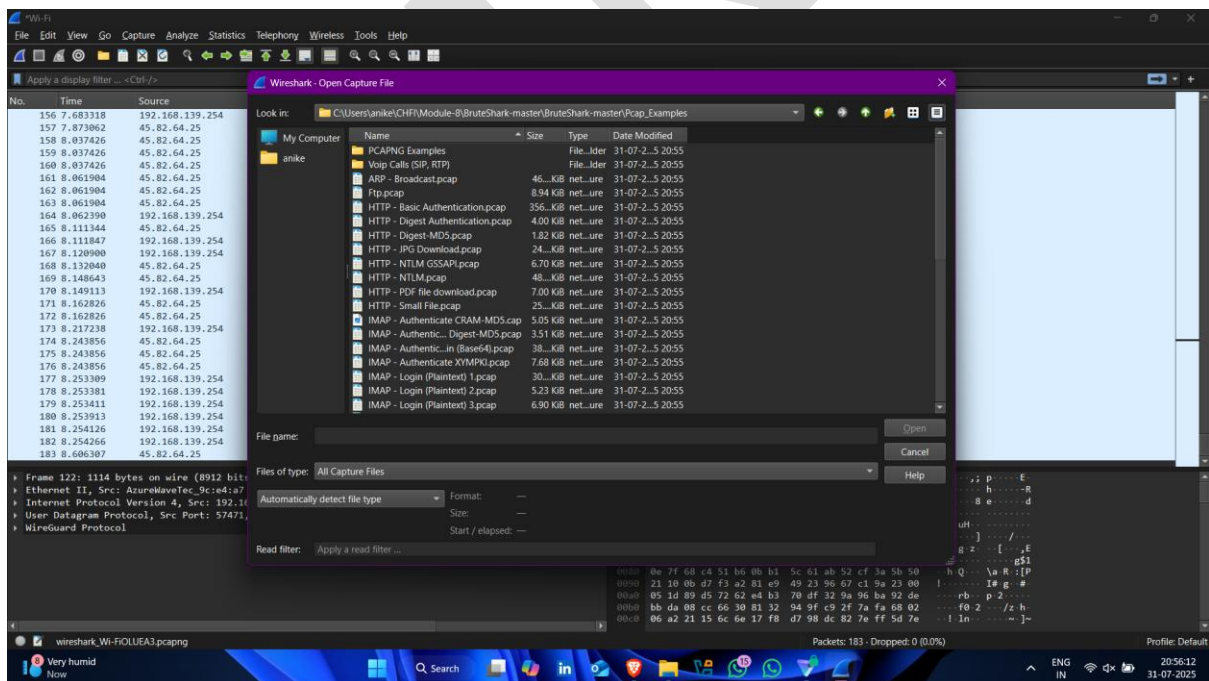
- Now click on **Open**



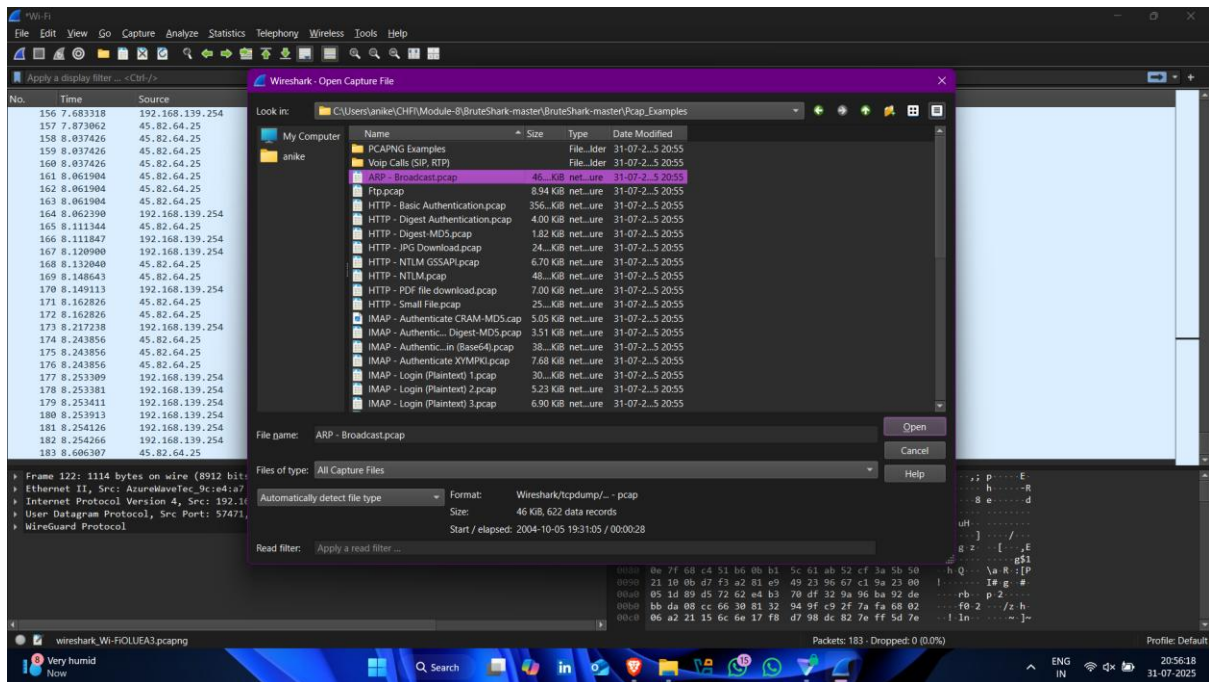
- Select log files



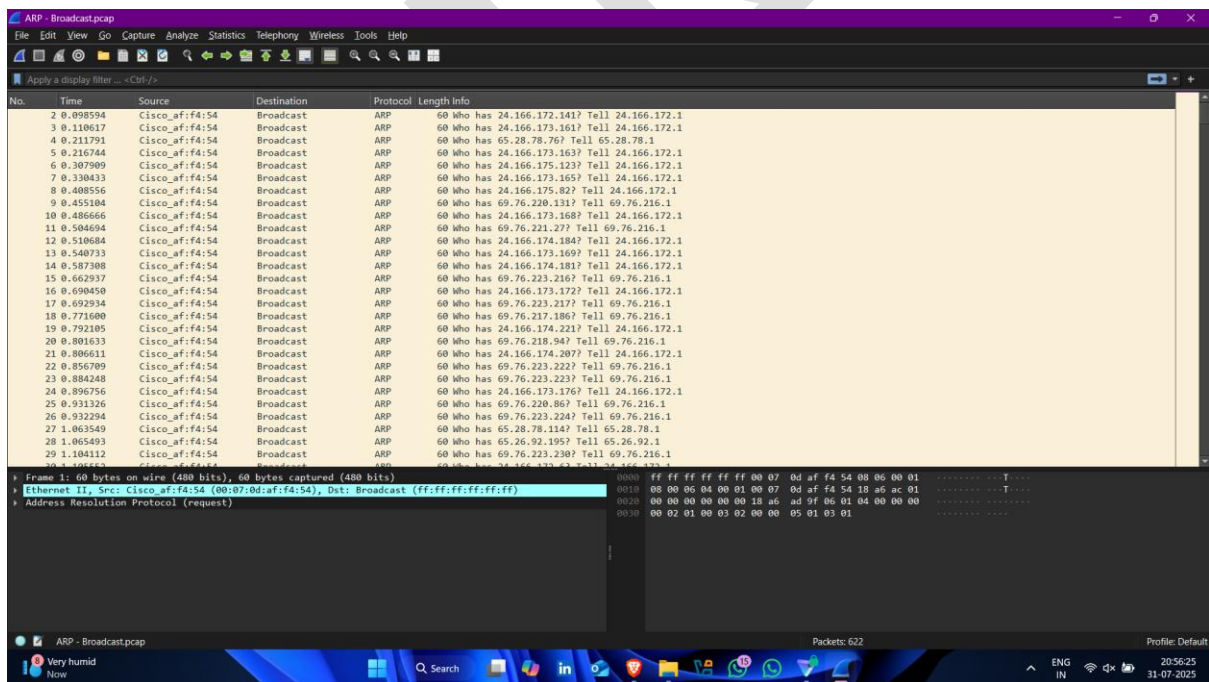
- select log



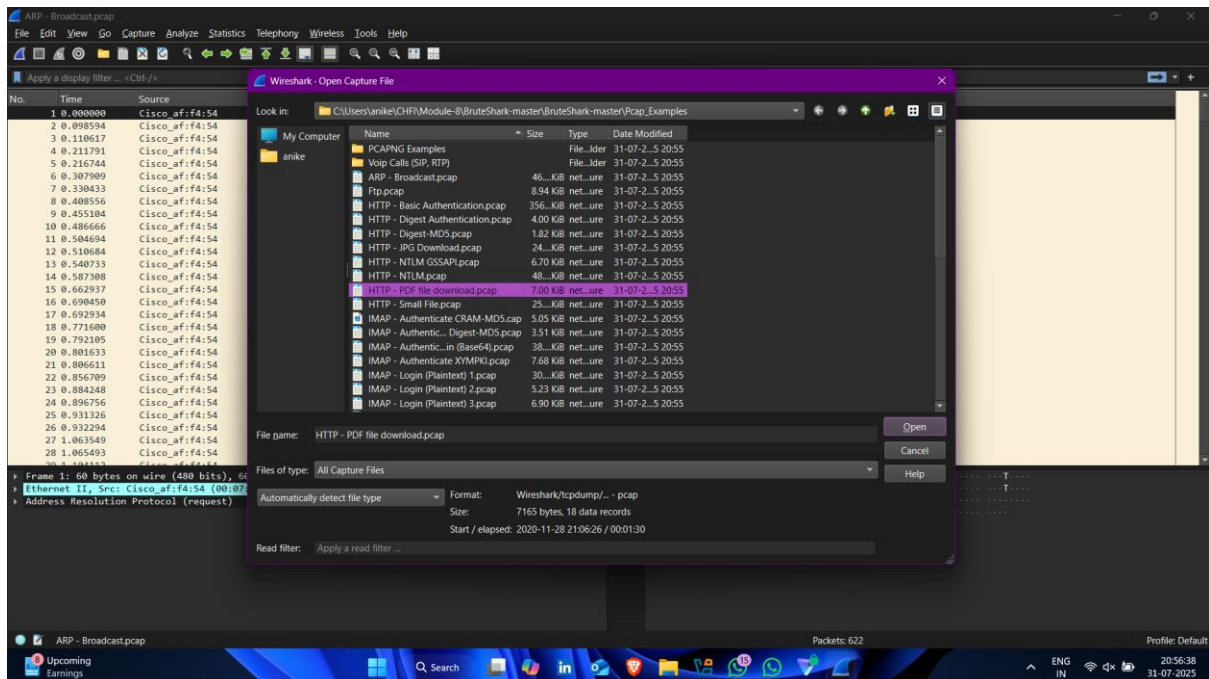
- click on open



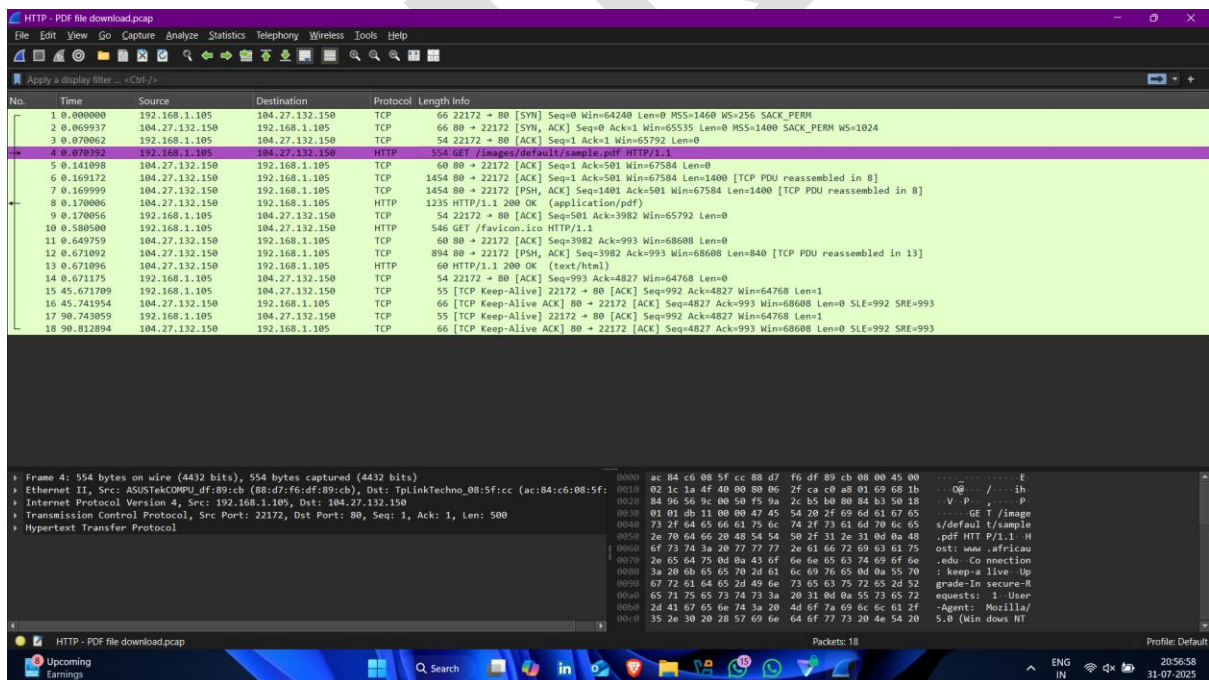
- Arp request packets



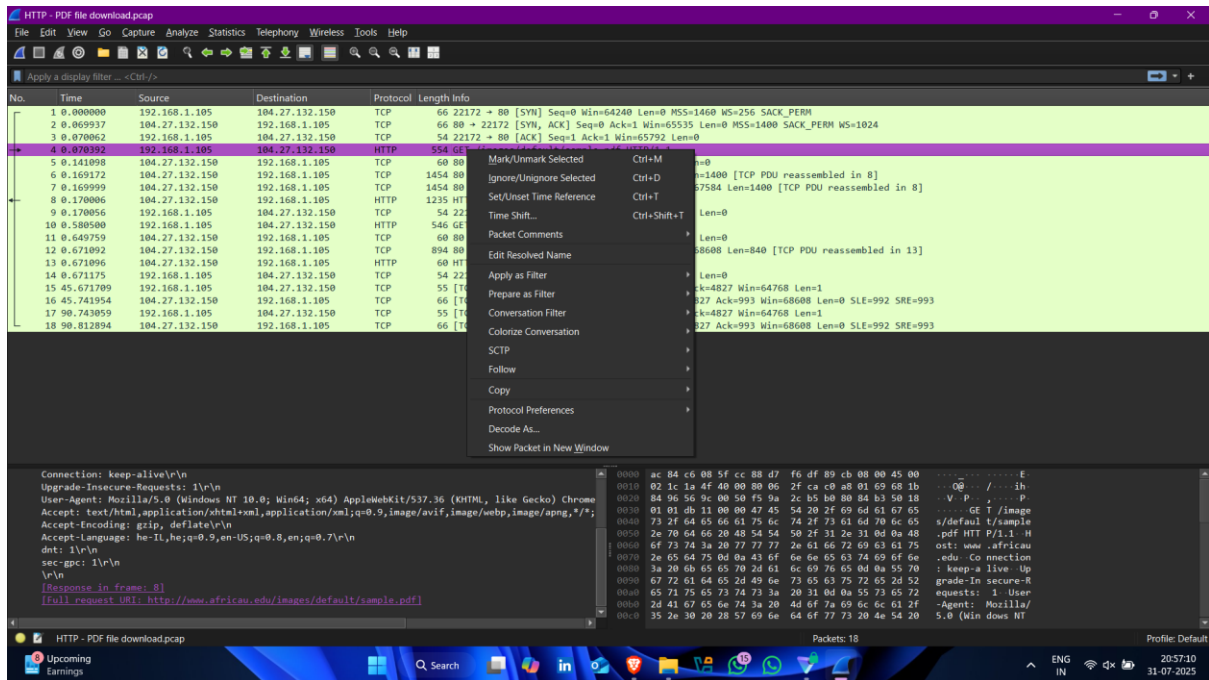
- Select Http log



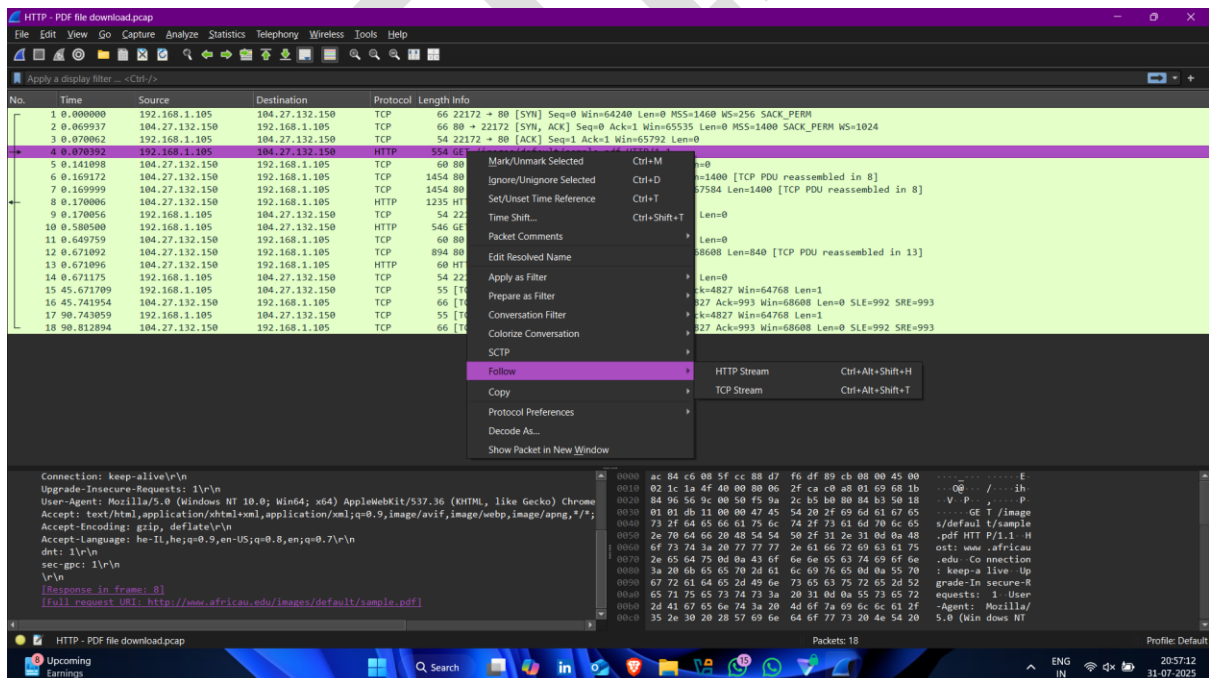
- Http packets log



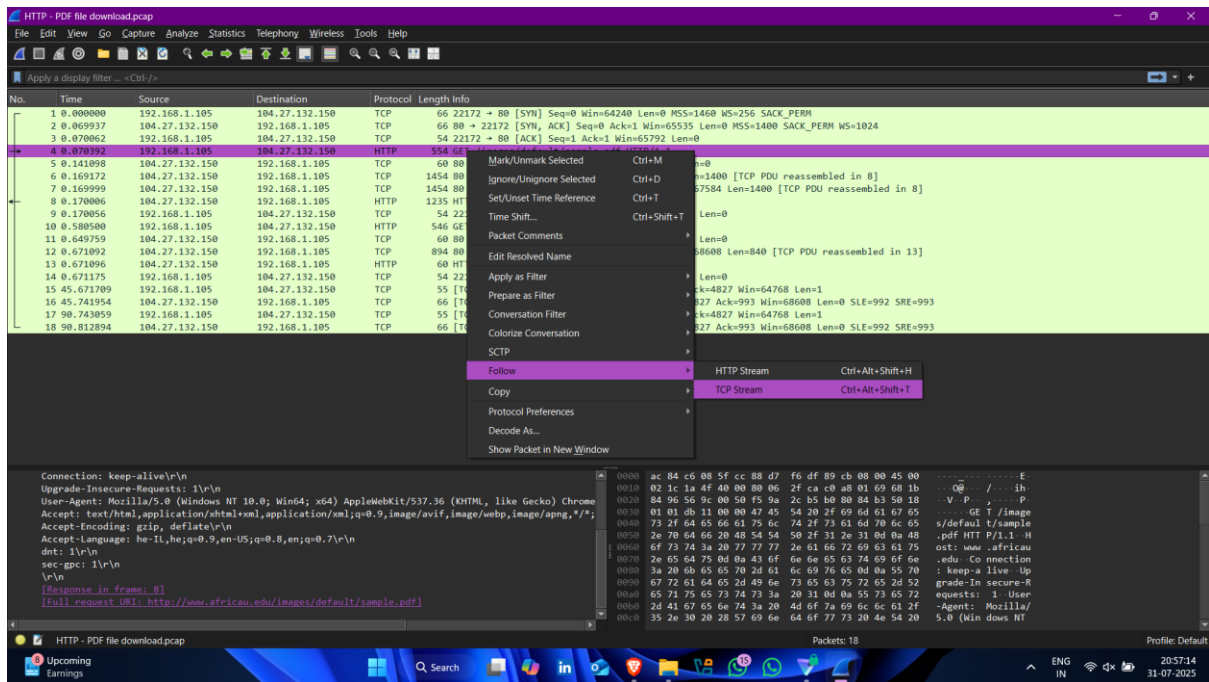
- Now Analyse packets
- Click on packet



- Click on follow



• Click on **TCP Stream**



• Packet Detailed

